

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

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Abstract

Faults in building HVAC equipment persist because the tools meant to catch them either raise alarms that operators learn to ignore, or flag anomalies that nobody can act on. STRATA turns ordinary one-minute trend data into an event log along the control sequence, discovers process models at the device and unit level, and scores each day through eight calibrated, physically interpretable channels against a measured noise floor, so that every alarm names a location, states its evidence and is held to a false-alarm budget. It was built under a falsification protocol: every hypothesis was pre-registered with the observation that would refute it (its falsifier), every result was recomputed independently before it was quoted, and automated tests pin the published numbers and the falsifiers that fired.

On five public LBNL benchmark systems, including the two fan-powered terminal units for which no detection results had been published before, STRATA detects 158 of 175 seeded faults, most on the first day, at 26 false-alarm days in 480 held-out fault-free days (1 to 9.4% per system). Two systems were onboarded blind, by configuration alone. The alarm names the correct terminal unit in 37 of the 50 detections with known ground truth and a wrong one in none, and the faulted subsystem in 83% of all detections. The false-alarm budget holds under injected sensor error, change-of-value logging and schedule shifts (worst 7 of 96 days) and on one measured rooftop unit (3 of 49).

Process mining helps to get detection but is not detecting on its own. The strata and the event abstraction carry the localisation, and invariants that discovery surfaced became several of the rules that detect. Conformance checking of the discovered models, pre-registered as the detector, was refuted: over 199 scenarios on six systems, no fault is caught by the models alone. A principal-component baseline detects three fewer of the 73 scenarios it shares and cannot name the device. A third-party Guideline 36 rule battery, held to the same gate, matches STRATA within one detection on the air handler and detects a third as many faults on the terminal units, so the pre-registered improvement claim over current practice is withdrawn there. The evaluation also produced eight machine-verified errata in the benchmark datasets. Every number regenerates from the public repository.

1 Introduction

Buildings and construction account for 32% of global final energy use and 34% of energy-related carbon dioxide emissions [74], and HVAC accounts for up to half of an individual building’s consumption [9] and for about 30% of all commercial building energy in the United States [25]. Energy wasted through faulty operation and poor control has been estimated at 15–30% of commercial buildings’ consumption [9, 39], and correcting the common faults at 5–30% whole-building savings [9, 63]. Until late 2023 these estimates rested on relatively small studies. The first large-scale

empirical characterisation appeared when Crowe and colleagues analysed multi-year fault-detection output across more than 60,000 pieces of HVAC equipment in 317 commercial buildings [16]. Their study reports an average of 245 faults per building per month and finds 40% of all air handling units in a faulted state on any given day. The most common AHU fault in their dataset, a supply-air-temperature setpoint that is never reset, was observed on 55% of the AHUs surveyed. These buildings were not unmonitored; many run modern building automation systems with alarms and fault-detection features, yet the faults persisted, which points to a gap in the kind of fault these tools can see rather than in their deployment.

The nature of that gap is specific. Of the ten most common AHU faults catalogued by Crowe and colleagues, seven are behaviour-based rather than condition-based: incorrect schedules, unreleased manual overrides, missed setpoint resets, and broken economiser sequences manifest as deviations in the temporal ordering of system states rather than as isolated component failures. A threshold alarm fires when a single variable leaves its acceptable band, and a pre-programmed rule fires when a named fault pattern is matched [66], but neither mechanism detects a control sequence that executes its steps in the wrong order or skips a step entirely unless a rule was written for that specific sequence in advance. That is possible, and has been done for logic faults [31], but only for the sequences someone anticipated. The structural argument that follows is straightforward: detection methods that model the sequence in which a building moves between operating states have a natural advantage on the fault classes that dominate the empirical landscape. The public benchmarks on which this paper is evaluated seed component, sensor and control-loop faults and none of that behaviour-based class, so the argument motivates the design and is not tested by it (Section 8).

The dominant research methodology for HVAC fault detection does not model that sequence either. Machine learning has been surveyed across 211 AI-based HVAC FDD studies published between 2013 and 2023 [9], with reported accuracies on the standard benchmarks saturating at 97–99% across hybrid random-forest pipelines on RP-1312 data [73] and convolutional-and-LSTM hybrids on real-building data [62]. The most recent work has shifted toward interpretability, but in a specific way: a deep model is paired with a post-hoc explanation method such as layer-wise relevance propagation or attention attribution [15, 19, 43, 83, 87]. The resulting explanation is at the level of input features rather than at the level of the control sequences the building was supposed to follow. A facility manager told that sensor X had high relevance for a prediction knows that something is unusual about that signal, but is not told how the building was supposed to be behaving instead.

A second problem concerns reproducibility. A 2025 audit of 65 primary machine-learning HVAC fault detection studies found two that shared code links, one of them broken, and that 72% do not specify whether their dataset is public, proprietary, or commercial [52]. The field has plateaued on accuracy, a metric that does not measure what an operator needs, with little of the existing work independently verifiable, and with the dominant explanation style answering a question (which signal mattered) different from the one operators ask (what should the system have been doing).

The gap, stated as an operator would state it, is a detector that works from the trend data a building automation system already records; that names the unit and the subsystem it implicates and explains the alarm in the physics of the equipment; that is held to a false-alarm rate measured on the building’s own fault-free days before its first alarm is trusted; and that is brought to a new building by configuration rather than by new code. Rule libraries supply the explanation and, on the terminal units measured in Section 6.4, a fraction of the coverage; anomaly detectors supply coverage but neither the name nor a budget; and neither is ordinarily evaluated so that a reader can check the claim. Process mining, the one methodology built to model the sequence in which a system moves between states, has not been brought to this problem: to our knowledge

no published work applies event-log process mining to building HVAC fault detection. The 211-study survey of AI-based HVAC fault detection contains no process-mining entry [9]; the reviews of process mining in Industry 4.0 (47 studies) and on sensor data (36 studies) contain no building HVAC application [3, 10]; the Lawrence Berkeley National Laboratory taxonomy of fault-detection methods has no such leaf [30]; and our own exact-phrase searches of the OpenAlex corpus return none (Section 3).

The opportunity is to change the model class. Process mining is a methodology in which a discrete event log derived from system observations is used to discover a Petri-net model of expected behaviour [54], with conformance checking against that model producing an alignment-based deviation signal that carries formal fitness and precision semantics [2, 75, 77]. We apply this approach to HVAC fault detection in stratified form, treating sensor-derived events as the input log and ASHRAE Guideline 36 control sequences [4] as the conceptual reference: events are abstracted from the control sequence at device and unit granularity, models are discovered per stratum, and the channels that read the resulting log (absence, frequency and conformance) sit beside the physics rules, residuals and oscillation channel that read the raw signals, each calibrated on its own noise. Detection is done by these calibrated, physically interpretable channels against a measured noise floor.

Our initial hypothesis was that the discovered Petri net would serve as both the expected-behaviour model and the detector. That hypothesis was pre-registered, tested, and refuted (Section 7): a single hand-written rule out-covered the unit-stratum model channel on the scenario where that channel was strongest. What the refutation settled is the division of labour, and it is a productive one. The process-mining structure carries the localisation and organises the detection: the event abstraction and the strata give every alarm its location, the absence and frequency channels read the log directly, and the frequency and oscillation channels are the only ones that catch the unstable-controller faults, and several of the rules that detect were written from invariants that discovery surfaced. Conformance checking of the discovered nets, the part we expected to detect, contributes no detection of its own. Process mining helps to get detection but is not detecting on its own. Whether the models can predict, in advance, where such a rule will hold on another building is a further claim this paper set out to test and could not (Section 7).

The paper makes four contributions.

1. The framework and the detector it yields. Eight calibrated channels over a stratified event log, reporting at a stated false-alarm budget rather than at accuracy alone. On five public benchmark systems, including the two fan-powered terminal units for which no detection results had been published before, the detector catches 158 of 175 seeded faults at 26 false-alarm days in 480 held-out fault-free days, names the faulted subsystem in 83% of detections and the correct terminal unit in 37 of 50 checkable detections with none wrong, and holds its budget under injected field conditions and on one measured rooftop unit; two of the five systems were onboarded blind, by configuration alone, and one alarm is shown verbatim (Section 6). Against an audited principal-component baseline it detects three more of the 73 scenarios they share; against a third-party Guideline 36 rule battery held to the same gate, it is within one detection on the air handler and detects three times as many faults on the terminal units, so the pre-registered improvement claim over current practice is withdrawn on the air handler and stands on the terminal units as a coverage difference (Section 6.4).
2. What process mining contributes to HVAC fault detection, established under pre-registration. This is, to our knowledge, the first application of event-log process mining to building HVAC fault detection, corroborated by the one systematic review of process mining on sensor data whose scope could have captured such work [10]. Its contribution is measured rather than

assumed: the event log, the strata and the invariants that discovery surfaced are what give the detector its location and several of its rules, and the frequency and oscillation channels are the only ones that see the unstable-controller faults; conformance checking of the discovered models, tested as the detector, catches no fault that the calibrated channels miss (Section 7). The same shape of result, that process models organise the evidence better than they detect on their own, has been reported for intrusion detection on network logs [90]. A further pre-registered test, of whether the models predict where a hand-written rule transfers between buildings, could not be given statistical power with the four buildings available and is reported as such rather than as a result in either direction.

3. The evaluation protocol. Pre-registration, honoured falsifiers, adversarial recomputation, tests that pin disclosures, and full regeneration of every committed result file (an artefact, in the sense used throughout) (Section 5), in a subfield where an audit of sixty-five primary studies found two that shared code links, one of them broken [52]. The protocol is applied to the paper’s own claims: what it refuted or withdrew is reported in Section 7.
4. Eight machine-verified errata in the benchmark datasets (Section 5.5), each with the command that reproduces it.

The paper proceeds as follows. Section 2 states the research aim and six objectives, each with what was done and where it stands. Section 3 places the work against machine-learning, process-mining and rule-based practice. Section 4 describes the event abstraction, the strata and the calibrated channels, and Section 5 the falsification protocol under which every result was produced, including the eight dataset errata it found (Section 5.5). Section 6 reports the results, including the two blind onboardings and the audit that mapped every recorded column; Section 7 what the evidence ruled out and what it settled; and Section 8 the limitations. Section 9 states what was found, what it is worth to an operator, what deploying it on the author’s campus would require, how it scales, and what a self-healing building would still need.

2 Research Aim and Objectives

This research set out to build a fault detector for building HVAC systems that an operator can trust and act on: one that works from the trend data the building already records, explains each alarm in the physics of the equipment, names the unit and the subsystem at fault, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining was the means chosen to get there, because it is the one methodology built to model the sequence in which a system moves between states, and no published work had applied it to this problem. The aim therefore carried a second question: what the discovered process models contribute to such a detector, and whether they detect on their own. Everything reported here is validated on public benchmark data; deployment on an operating building is the intended next phase, and Section 9 states what it would take.

What the framework is meant to do. A building operator today has two ways of finding a fault: wait for an occupant to complain, or watch the trend graphs and try to spot by eye a curve that has drifted from where it usually sits. Both find the fault late, both depend on someone happening to look, and neither says which piece of equipment is responsible. That is the situation this framework is meant to change. STRATA is meant to consume the building’s ordinary one-minute trend data and one configuration file that names its sensors and control rules, learn from a fault-free period

what healthy operation looks like, and then, every day, name the days that break a physical relationship the healthy period established: which unit, which relationship, by how much, and how often that channel has cried wolf on healthy days. The output is a work order, not a graph, with a false-alarm budget the operator can hold the system to. Onboarding a new equipment type should be a configuration exercise, not a software project. That intent fixes the six objectives below, each of which is reported with what was actually done and where it now stands.

1. Abstract events from sensor data. Objective: convert continuous HVAC sensor streams into discrete event logs using control-sequence rules grounded in ASHRAE Guideline 36, with fault-signature events held out of model discovery so that the models cannot re-detect the rules that produced them. Done: an event-abstraction layer driven entirely by configuration: each day is a case, and every event is emitted by a rule in the building’s configuration, of eight kinds, three of which (occupancy, operating mode, time window) produce the state events that describe what the equipment is doing and five of which (command-against-measurement mismatch, leak, setpoint deviation, paired residual, envelope residual) produce the fault-signature events. The same code, with no building-specific additions, ran on seven LBNL datasets (five simulated systems, a simulated and a measured rooftop unit); the three residual operators added during the work were added to the library under pre-registration and apply to every system. The alphabet wall between state and fault-signature events is enforced by unit tests, and the circularity it prevents was measured rather than asserted (Section 4). Status: met.
2. Discover models and locate invariants. Objective: learn Petri-net models of expected control behaviour from fault-free data at device, unit and cross-system granularity, and establish what those models contribute. Done: nets were discovered at all three strata on every system, and their contribution was measured under pre-registration: conformance against them is recorded as contributing no detection that the calibrated channels miss, on 199 scenarios across six systems (a one-sided 95% upper bound of 1.5% on the sole-detection rate), with a positive control showing the channel can see order on two systems, a seven-instrument grid showing the null is not the miner’s, and every recorded column finally read. Their contribution was at design time: invariants they surfaced became rules that detect (Section 7). Status: met. The models’ contribution is established and stated: the structure, the event log and the invariants they surfaced at design time. The hypothesis that they would also detect by conformance alone is refuted.
3. Detect against a measured noise floor. Objective: calibrate interpretable channels on fault-free data alone and require every alarm to exceed that channel’s own measured noise, so that detection is reported with a stated false-positive budget rather than with accuracy alone. Done: eight channels, every threshold from the fault-free year, a three-way calendar split so that every false-alarm rate is out-of-sample, and a per-day noise floor per channel; the budget held under injected field conditions and on one measured building (3 of 49 held-out days). Status: met; the cost is stated where it bit (the statistical repair removed seven detections, Section 8).
4. Benchmark against known answers. Objective: evaluate on the LBNL benchmark family against documented ground-truth faults, and compare with a machine-learning baseline on identical sets of scenarios and with a third-party implementation of the ASHRAE Guideline 36 rule battery. Done: 158 of 175 seeded scenarios detected on the five simulated systems (14 of 14 with 13 adjudicated, one detection having been attributed on audit to a dataset defect

rather than to the fault; 26 of 30, 25 of 29, 50 of 55, 43 of 47; 157 under the adjudication) at 26 false-alarm days in 480, most on the first day. Against an audited principal-component baseline on identical universes, 64 of 73 to its 61, with three scenarios only STRATA catches and none only the baseline catches. The seventeen remaining misses are coil fouling whose recorded values sit inside every healthy band, and one is visible but uncertifiable from a single healthy year (Section 6.10). Done, against the claim: the pre-registered comparison with the open-fdd implementation of the Guideline 36 battery (Section 6.4) fired its own falsifier: held to the treatment STRATA gives its own signature rules, the battery detects 12 of 14 on the single-duct air handler at one false-alarm day against STRATA’s 13 at one, and 8 of 30 and 9 of 29 on the terminal units at no false-alarm day against STRATA’s 26 and 25 at eight and nine. Status: all three parts executed; the improvement claim over current practice is withdrawn on the air handler and stands on the terminal units only as a coverage difference bought with false alarms.

5. Make every claim checkable. Objective: pre-register each hypothesis with the condition that would refute it, subject every result to adversarial recomputation before it is quoted, and guard published values, fired falsifiers and disclosure obligations with automated tests. Done: more than thirty pre-registered experiments (Appendix A), ten adverse results honoured and published together with four later fired falsifiers reported beside the results they bound, every headline number regenerated from a committed artefact under a regeneration gate, and a test suite that fails the build if a published caveat is dropped. The protocol caught our own errors: a drift between the library and the scripts, a statistical floor that was looser than the one it replaced, and sixteen “unreachable” misses that were columns nobody had mapped. Status: met; this is the contribution the paper is surest of.
6. Release the work openly. Objective: publish the pipeline and its building configurations as open-source software, evaluated on public data with a documented path from a fresh clone to regenerated artefacts. Done: the repository, every configuration, every artefact and the reproduction path are public; eight machine-verified defects in the benchmark datasets are published with reproducing commands (Section 5.5). Status: met.

What the objectives add up to: a detector that does what the intent above asks on the benchmarks it was tested on, a settled and well-defended answer to the process-mining question that motivated it, and a protocol that made both claims checkable. What they do not yet add up to is a field result: the measured building contributes a false-alarm rate, and its single fault case was not visible in its recorded signals. That result is the next phase, and Section 9 states what it would take on the author’s campus: one fault-free season of trend data, a sensor map, and a pass confirming that the rules are silent on that fault-free period.

One goal lies outside this scope and is stated as future work rather than quietly dropped: deployment on an operating building, which needs a healthy period to learn from and the sensors mapped. The coverage audit of Section 6.10 is the checklist that deployment would start with, and Section 9 states what deploying on the author’s campus would require and deliver.

3 Related Work

3.1 Machine learning for HVAC fault detection, and the move from accuracy to interpretability

The machine-learning era of HVAC fault detection is well documented. Bi and colleagues survey 211 AI-based HVAC FDD studies published between 2013 and 2023, organising them into traditional machine-learning, deep-learning, hybrid AI, and physical-model branches [9]. Reported accuracies on the standard benchmark datasets have effectively saturated. Tun and colleagues reached 98% accuracy across thirteen fault types using a hybrid random-forest and support-vector-machine pipeline on RP-1312 AHU data [73]. Prince, Yoon, and Kumar reached 97% accuracy on data from three real buildings using a thirteen-layer one-dimensional convolutional-and-LSTM hybrid that remains robust to Gaussian noise and to 10% missing values [62]. The authors of that paper close by explicitly identifying interpretability as the problem they did not solve, proposing grafting SHAP or LIME as future work.

The most directly relevant interpretable benchmark for the LBNL SDAHU subset is Deng and colleagues’ adaptive fusion graph convolutional network. The model reaches 88.09% accuracy at 500 labelled samples and 75.15% at only seven samples, with per-node sensor importance as the explanation mechanism [19]. Subsequent work in the same vein pairs multi-head attention with parallel dilated convolutions and applies layer-wise relevance propagation for explanation on chiller fault data, reaching an F1 score of 0.9841 across 27 operating conditions [83]. The pattern across this body of work is consistent: a deep black-box network is the model, a post-hoc analysis layer provides the explanation, and the explanation is framed at the level of input features rather than control flow. The architectural opportunity is for a method whose model is its explanation.

3.2 Unsupervised methods and cross-building deployment

Real buildings rarely come with ground-truth fault labels, and the strongest recent work on unlabelled data uses self-supervised or autoencoder methods. Abdollah and colleagues pretrained a Transformer encoder via masked-token reconstruction on unlabelled HVAC time-series from a heat-pump building at the Polytechnic University of Milan, then used Peak-Over-Threshold detection on reconstruction error to flag a real two-month monitoring fault between October and December [1]. The most directly relevant precedent for a Canadian-university real-BAS deployment is El Mokhtari and McArthur’s autoencoder-based work at Toronto Metropolitan University, which distinguishes equipment-level from system-level faults across five fan coil units and demonstrates cross-unit transfer learning [20]. These methods share a common limitation. Each produces a learned latent embedding whose deviation signal is a scalar anomaly score, not a process-level explanation.

The cross-building generalisation problem is the binding constraint that prevents most existing HVAC FDD work from being deployed in the field. The established response is domain adaptation, which transfers learned network weights [21,42]: Zhu and colleagues migrate fault detection between two different building chillers [91], and Ghalamsiah and colleagues apply a contrastive adaptation network to ASHRAE-1312 simulation as a labelled source and real AHU measurements as an unlabelled target [23]. What STRATA transfers between buildings is a configuration file naming sensors and rules, not a set of learned weights; transferring it is therefore a different operation from the weight transfer the literature above studies, and Section 7 reports that the discovered models themselves did not transfer.

3.3 Process mining: foundations, adjacent successes, and the empty quadrant in building energy

Process mining was formalised by van der Aalst as a methodology for discovering process models from event logs, checking observed behaviour for conformance, and enhancing the resulting models with additional perspectives [75, 76]. Conformance checking was used directly as a trace-level anomaly detector on event logs [8, 64]; the line subsequently moved to neural sequence models that outperform conformance-based baselines on standard log corpora [56]. The conformance channels used here are the nearest descendant of the former, and the ablation reported in Section 6, which finds no fault detected only by them, is consistent with the motivation for the latter. The methodology has also been applied across cyber-physical and sensor-driven domains adjacent to building energy [55, 68]. Vitale and colleagues applied a full process-mining pipeline to fault diagnosis on the RoAD robotic-arm dataset, reaching an F1 score of up to 98.925% with a conformance-checking time as low as 0.020 seconds [81]. The same group then demonstrated that combining process-mining-derived alignment features with downstream dimensionality reduction substantially outperforms pure conformance checking across four cross-domain datasets, with F1 improvements ranging from 30 to 70 points [82]. Husom and colleagues’ UDAVA pipeline extended this hybrid pattern to industrial Internet-of-Things sensor validation [34]. Methodologically the closest analogue to building HVAC is Moradbeikie and colleagues’ work on pasteurisation, where continuous multivariate sensor signals are abstracted into discrete activity sequences via hidden Markov models and conformance is checked against HACCP critical control points [50]. Pasteurisation, like HVAC operation, is a continuous-flow process governed by an explicit regulatory standard; the analogy between HACCP critical control points and ASHRAE Guideline 36 control sequences is suggestive. It is not a direct port: pasteurisation has a single physical flow and a case notion given by the batch, whereas HVAC operation has no natural case boundary, concurrent devices at several granularities, and an alphabet in which some events already encode fault suspicion. Both the case notion used here (the calendar day) and the partition of the event alphabet had to be constructed. The technical barrier common to all these applications is event abstraction, and it remains active: the newest work from the authors of the sensor-data review is a method for extracting event logs from raw industrial sensor data [11]. Event abstraction, the conversion of continuous sensor time-series into discrete activity sequences [17, 47], which was formalised in the methodology paper by Janssen and colleagues [36].

Despite these adjacent successes, building HVAC remains an empty quadrant in the literature. The novelty claim rests principally on one review and is corroborated by three others whose scope or period would not necessarily have captured such work. Bi and colleagues’ survey of 211 AI-based HVAC FDD studies contains no process-mining entries in any leaf of its method taxonomy [9]. Akhramovich and colleagues’ review of 47 process-mining studies in Industry 4.0 covers manufacturing, internal logistics, mining, alarm management, supply-chain logistics, and product lifecycle management; building applications do not appear [3]. Where process mining has reached the built environment at all it is for construction and facility-management workflows [48], with cases as projects and activities as human tasks; the operational control flow of a building’s mechanical systems is not addressed. Brzychczy and colleagues’ more recent review of 36 process-mining studies on sensor data documents the same absence at finer granularity: its applications are industrial, smart-home (motion-sensor occupant-activity recognition), healthcare, smart-office and smart-product, with no building HVAC among them [10]. The Lawrence Berkeley National Laboratory team’s canonical FDD-methods taxonomy from 2017 organises automated fault-detection methods into quantitative model-based, qualitative model-based, and process-history-based categories, and process mining is not present at any leaf [30]. The absence is consistent across all four, and our own

exact-phrase searches of the OpenAlex corpus [61] for process mining or conformance checking in conjunction with HVAC, building automation, air handling, or chillers return no work; nor did the ICPM 2025 programme, the most recent at the time of writing, contain any. No published work, to our knowledge, applies event-log process mining to building HVAC fault detection.

3.4 Rule-based fault detection in practice, and what process mining adds

Fault detection is not absent from real buildings; it is simply rule-based. The most widely deployed form is the fault-detection capability embedded in commercial building automation systems. Modern platforms ship libraries of pre-programmed fault rules: the Automated Logic WebCTRL system at the Campus Activity Centre of Thompson Rivers University, the building intended for this work’s deployment phase, for example, includes a standard library that matches over one hundred named faults in common HVAC equipment.¹ The characterisation survey of fourteen commercial FDD tools by Granderson and colleagues confirms that this rule-based and threshold-based style is the norm across the commercial landscape [30], and Guideline 36’s fault conditions have themselves been used as the reference against which learned diagnostic models are built [60]. The strength of a rule library is that it works immediately on connection and requires no baseline data; its structural limitation is that it can only detect the fault patterns that were coded into it in advance, and so it misses novel failure modes and the gradual, multi-variable, sequence-level deviations that no rule author anticipated. Section 6.4 measures both the strength and the limitation directly, against a third-party implementation of the Guideline 36 battery on the same benchmarks.

The same rule-based philosophy characterises the most directly comparable research programme, the Lawrence Berkeley National Laboratory team’s automated fault-correction line [44, 45]. Pritoni and colleagues field-tested seven hand-coded correction algorithms across 225 distinct equipment items in four buildings on three building-automation-system platforms, targeting common control problems such as incorrectly programmed schedules, unreleased manual overrides, and missed set-point resets [63]. Each algorithm is a pre-encoded rule matched against BAS data and applied through a two-way BACnet writeback to the building’s control system. Both the commercial libraries and the LBNL correction algorithms share the assumption that the faults of interest can be enumerated and encoded ahead of time.

STRATA makes the opposite assumption and is therefore complementary to both. Rather than matching observed data against a fixed catalogue of known faults, it learns from the building’s own fault-free operating data where that building’s behaviour is invariant, and calibrates detection channels to hold those invariants. Conformance alignment against the discovered models is one such channel rather than the whole detector, a distinction our own results forced on us (Section 7), and it contributes the property a rule library cannot supply unaided: during development the invariants were surfaced by discovery rather than enumerated in advance, which is how several of the rules now deployed came to be written. Every channel carries a measured false-alarm budget, which is what makes the resulting alarms answerable rather than merely numerous. A practical consequence is that STRATA can run on the same trend data a commercial system such as WebCTRL already records, addressing faults that fall outside the rule library or below its thresholds, while the rule library continues to handle the well-understood patterns it was built for. On the benchmarks used here those are the terminal-unit faults that Section 6.4 measures; the behaviour-based faults the introduction cites are not seeded in any of them (Section 8). In principle the located, evidenced alarms STRATA produces could serve as input to the LBNL

¹Automated Logic, Fault Detection and Diagnostics, <https://www.automatedlogic.com/en/products/webctrl-building-automation-system/integration/fdd-reporting-and-dashboards/>, accessed 2026-09-28; the page states that the library “can pinpoint over 100 proven faults in typical HVAC equipment”.

correction algorithms, extending automated correction beyond the patterns pre-encoded by hand. The LBNL FDD dataset family [27–29] provides the labelled simulation testbed that supports this complementary positioning.

One adjacent body of work must be distinguished as a class, because its vocabulary is similar. Hand-designed Petri nets have modelled HVAC and building systems for two decades: for hybrid system modelling [80], for building-automation monitoring [26], and for chiller fault diagnosis [88]. In every case the net is authored by an engineer rather than discovered from a log. The most recent example is control: Deabes, Bouazza, and Alghami designed a fuzzy Petri-net controller for HVAC temperature regulation, hand-engineering both the Petri-net structure and the fuzzy rules and using the model as a forward control device for an air-conditioning compressor [18]. None of this work contains an event log, a process-discovery algorithm, or alignment-based conformance checking; the diagnosis variant matches observations against a net an expert drew. The present work does the methodological opposite: a Petri net is discovered from a real event log of HVAC operation, then used as a reference against which alignment-based conformance checking diagnoses operational deviations. Hand-designed Petri-net controllers and data-discovered Petri-net diagnostic models are two distinct uses of the Petri-net formalism, and they should not be conflated.

4 Method

STRATA (Figure 1; code and artefacts at <https://github.com/iyassh/strata>) consumes one fault-free year of ordinary building-management-system trend data together with a single configuration file naming the building’s sensors and control rules. That configuration file is the only site-specific artefact: no code change was required to onboard any system reported here.

The framework is stratified in two senses. Events are abstracted from the control sequence at device and unit granularity, so that every alarm is born with a location; and the channels that read the resulting log (absence, frequency and the two conformance channels) sit beside the physics rules, the residuals and the oscillation channel, which read the raw signals, each calibrated against its own noise. The process-mining structure is what localises and organises detection; conformance against the discovered nets is one channel among eight, and Section 6 measures what it contributes.

4.1 Event abstraction and the alphabet wall

Continuous sensor streams are converted into a discrete event log using control-sequence rules grounded in ASHRAE Guideline 36 [4], following the abstraction methodology of Janssen and colleagues [36] and the taxonomy of event abstraction in van Zelst and colleagues [78]; the same bridge from continuous sensor streams to process-aware event logs is pursued, for IoT data generally, by Sensor2EventLog [51]. The events are then partitioned into two alphabets that are never mixed. State events are neutral facts about operation and are the only events admitted to model discovery. Fault-signature events already encode engineering suspicion and are withheld from discovery entirely.

The partition, the alphabet wall, exists because of circularity, the process-mining form of the data-leakage problem [38, 85], whose closest published relative is a benchmark-validity audit in sensor-based fault diagnosis that found reported accuracy re-encoding the labelling rules [65]: a model discovered over an alphabet that already names faults will later “detect” those faults by recognising the project’s own rules. Table 1 measures the effect on the single-duct system by comparing the committed artefact of the pre-partition prototype with the partitioned pipeline’s artefact under the same threshold, the same 248-day train and 87-day holdout split and the same 4,540 fault days (but not the same code version). Without the wall the model channel’s apparent

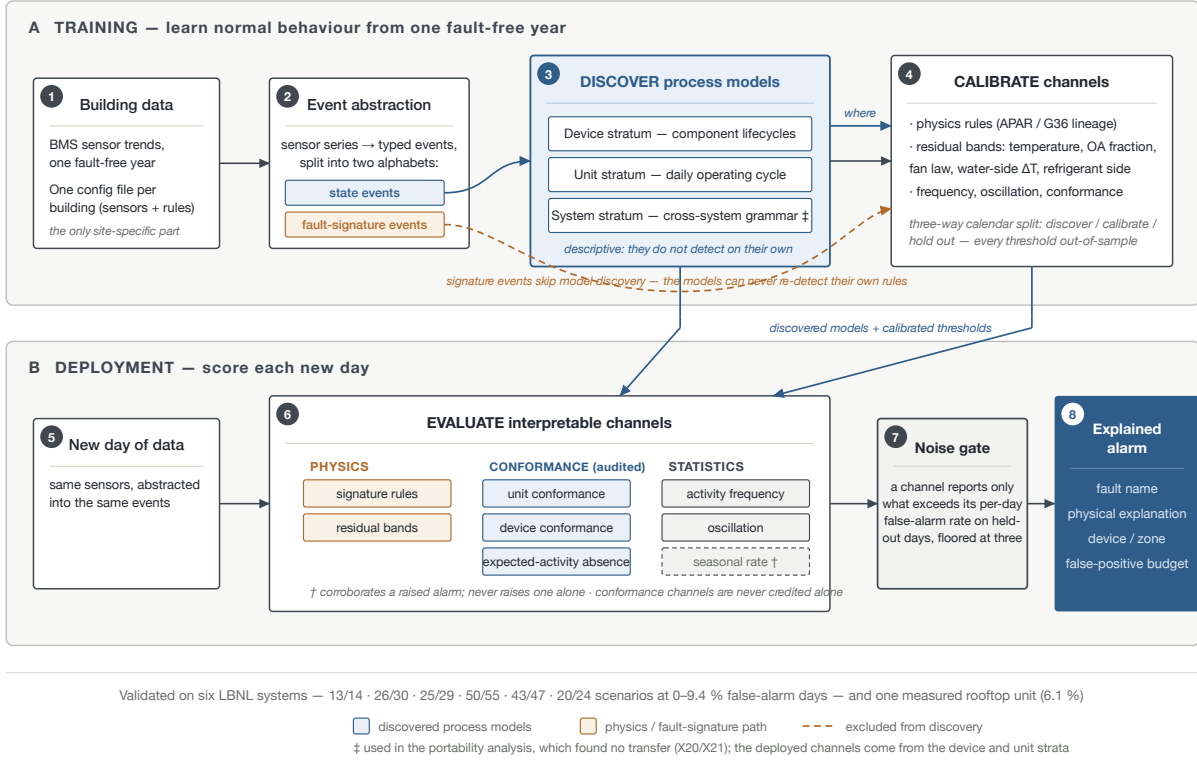


Figure 1: The framework as finally evaluated. Training (top): one fault-free year and one configuration file; events are split into state and fault-signature alphabets; process models are discovered from state events only and, on the evidence of Sections 6 and 6.8, describe normal operation and supply the strata and invariants the channels use, but contribute no detection of their own through conformance; every channel’s threshold comes from a three-way calendar split (discover, calibrate, hold out), so every reported false-alarm rate is out-of-sample. Deployment (bottom): each day is scored by physics, statistical and conformance channels; a channel reports only what exceeds its own per-day false-alarm rate on the held-out days it could evaluate, floored at three days; what survives reaches the operator with a name, a physical explanation, the device, and the stated budget.

recall is 5.2%; with it, 1.8%. Every combined-detector number is identical, because every model-channel detection the leak supplied was already caught by the rules channel: the partition changed the deployed detector’s output by zero days. That 3.5-percentage-point difference is a measured bound on model self-confirmation. A controlled same-code re-run is future work; the deployed detector’s committed results file (the scorecard, the term used throughout), two code generations later, gives the same reading (86 fault days strictly worse than the worst fault-free holdout day, of 4,891 evaluable fault days, 1.76%; the thresholded model channel flags none). The separation is now enforced by a tested invariant: injecting signature events into a log cannot change a day’s fitness. The process-mining literature warns about this trap in the abstract; we are not aware of a prior quantification of it on sensor-derived logs.

Table 1: The alphabet wall, measured on the single-duct system. Same threshold, same 248-day train and 87-day holdout split, same 4,540 fault days; the two artefacts differ in code version. The model channel’s apparent recall without the wall is self-confirmation; the deployed detector’s output does not change.

	without the wall (prototype)	with the wall (partitioned)
Rules channel, true-positive fault days	3,134	3,134
Model channel, fault days flagged (apparent recall)	238 (5.24%)	81 (1.78%)
Combined detector, true-positive fault days	3,152	3,152
Combined detector, false-positive holdout days	5	5

4.2 Discovery at three strata

Petri nets are discovered from the state-event log with the inductive miner [40, 41], as implemented in pm4py [7], at three granularities: the device stratum models an individual actuator’s lifecycle, the unit stratum models the air handler’s daily operating cycle, and a cross-system grammar supports the transfer analysis of Section 7.

Whether a discovered net encodes temporal order at all, rather than merely a set of permitted activities, is the question precision metrics address [12, 53, 70], and the imprecision of precision on logs without negative examples [79] is why we use a direct probe instead. The reversal probe is that direct check: every trace is reversed and re-scored, and the fitness drop is reported as a single number. The three models separate cleanly. The single-duct net falls from 0.8861 to 0.5526 under reversal, confirming genuine order sensitivity. The series fan-powered net scores identically to four decimal places (0.9533 against 0.9533) and the parallel net shifts by 0.0005 (0.9021 against 0.9026), showing that neither carries meaningful ordering information in its fitness. The probe is cheap, decisive and reusable, and it converted an informal claim about alphabet complexity into a measured fact.

4.3 Calibrated channels and the noise gate

Eight channels (Table 2) evaluate each day independently: (1) rules, physics rules descended from the NIST APAR lineage [66] that name direct contradictions between measurements; (2) residual, physical-residual bands encoding relationships that must hold under known operating modes (with the cooling coil inactive, supply-air minus mixed-air temperature should equal the supply-fan heat rise, so a biased sensor appears as a persistent offset); (3) model conformance and (4) device conformance, alignment cost [13, 77] against the discovered unit and device models respectively, the two process-mining channels; (5) absence, a scheduled activity that did not occur; (6) frequency and (7) oscillation, statistical bands on activity counts and reversals; and (8) a seasonal rate channel that corroborates but may not raise an alarm alone, for reasons disclosed in Section 8.

Calibration uses the fault-free year only, split three ways by calendar. The last eight days of each month (96 days) are held out and report every false-alarm rate; the eight days before each holdout block form a calibration slice from which the two conformance thresholds are taken, at a false-positive quantile of 0.01 on per-day fitness; the band channels (residual, frequency, oscillation) learn their extremes on every non-holdout day, and process discovery uses the days that remain. No threshold may see fault data, and no threshold sees the days its false-alarm rate is reported on. Every rule additionally requires a positive control, because silence is not competence: a position-based leak rule proved structurally deaf on the fan-powered units, where the seeded leak passes water through a valve whose position reads closed honestly, and was replaced by a waterside flow rule.

Table 2: The eight channels as deployed. Every threshold is set from the fault-free year alone: bands from its training days, conformance thresholds from its calibration slice, and the noise floor from its held-out days. “Alphabet” names which side of the wall the channel reads. The rate channel is advisory: reported, never counted in the detector’s alarm.

Channel	Alphabet	A day is flagged when	Threshold source	In union
Rules	signature	a control-sequence rule (Guideline 36 / APAR lineage) fires during operation	fixed bands from the sequence documentation; no fitted parameter, but a rule that is not silent on the fault-free year is removed at onboarding; gate: 3 of 365 (Section 6.4)	yes
Residual	raw signals	a per-mode physical residual (coil-off supply-mixed rise, outdoor-air fraction, static $\div$ speed ² , refrigerant approach) leaves its healthy band	[min, max] of training-day scores, widened to at least a per-rule sensor floor (0.5 °F for temperature residuals, 0.05 or 0.01 for flow fractions and ratios); exceedance must also clear that floor	yes
Model conformance	state	the day’s trace aligns to the unit net with fitness below threshold	1 % quantile of per-day fitness on the calibration slice	yes
Device conformance	state	an actuator’s daily lifecycle trace aligns to its device net below threshold	as above, per device net	yes
Absence	state	a device that is active on scheduled training days produces no lifecycle event on a scheduled day	each device’s healthy silence rate on training days	yes
Frequency	state	a daily activity count (unit or device) leaves its healthy range	[min, max] over training days, for activities seen on ≥ 5 % of them	yes
Oscillation	raw signals	an actuator’s daily count of direction reversals leaves its healthy range	[min, max] over training days	yes
Rate	state	a 30-day active-day rate leaves the range of healthy windows ending in the same calendar month	training-year windows, month-matched	advisory

A channel may then report only what exceeds its own measured noise floor, tested as a binomial at $p < 10^{-3}$. For the residual, conformance, absence, frequency and oscillation channels the null is that channel’s own false-alarm rate on the fault-free holdout days it could evaluate, floored at three days [32], a per-day rate since the statistical repair disclosed in Section 8. For the rules channel, whose rules are silent on the whole fault-free year by construction (a rule that fires there is removed at onboarding), the null is the same rule of three applied to the year, 3 of 365. Detection thresholds are additionally floored at sensor physics: an early result on the series unit, 21 to 117 detection-days per scenario, was retracted to at most two when a 0.5°F sensor-precision floor revealed that the “detections” had exceeded a 0.22°F band by 0.02 to 0.03°F .

What survives the gate reaches the operator with the implicated device or zone, the evidence that triggered it, and the channel’s own false-alarm budget. Figure 4, in Section 6, is one such alarm from the first day of a seeded fault, every value read from the committed artefacts and the raw data.

5 The Falsification Protocol

An audit of sixty-five primary machine-learning HVAC fault-detection studies found two that shared code links, one of them broken, and 72% that did not state whether their data was public, proprietary or commercial [52]. A field in that condition needs results that can be checked more than it needs another accuracy figure. We therefore treated verification as a deliverable, and this section describes the protocol in enough detail to be adopted: what is written down before a result is computed, what is done before a number is quoted, what is guarded afterwards, and what the protocol taught. The refutations it produced are reported in Section 7.

5.1 Before a result: pre-registration and gates

Every experiment is pre-registered. The hypothesis and the specific observation that would refute it are committed to version control before the experiment runs [57], in the spirit of the hypothesis-testing discipline recently urged for process-mining research [59], of the pre-registration template proposed for predictive modelling [33], and of the pre-registered protocols the simulation-study literature recommends against spurious claims of superiority [58]. A falsifier that fires is a finding and is published, not a setback to be absorbed by rewriting the hypothesis. Ten results went against the project: four refuted hypotheses, one fault-informed rule whose falsifier fired, three withdrawn claims, one pre-registered test that never had attainable power, and one four-building transfer test whose pass a discovery-free control reproduced. Four later pre-registered tests (component attribution, the diagnosis layer, quantile bands and the Guideline 36 comparison) also fired their falsifiers, and each is reported beside the result it bounds (Sections 6, 8 and 6.4). Appendix A lists every pre-registered experiment with its prediction and outcome. The discipline exists because we watched ourselves overclaim: an early phase report drafted on the day of its result contained six overstatements that audit caught within hours. Positive results attract inflated framing faster than negative ones attract scrutiny, and a falsifier only protects a claim if it is written down before the result it would kill is known.

Before any detection result is computed, a battery of gates runs against the raw data. It verifies file integrity by MD5, checks that timestamps increase monotonically, confirms that each file’s calendar is internally consistent, checks the raw files for calendar rotation, checks coverage of the Brick semantic model [5], and hashes each derived event log so that duplicate scenarios cannot masquerade as independent evidence. A separate healthy-silence gate runs per configuration and confirms that the rule set is silent on the fault-free year before it may score anything. None of

these gates was designed in advance; each was added after a dataset defect got past the previous set. Three of the eight errata in Section 5.5 were found by this battery (E1, E2 and E6 by the integrity hash); the others were found by audit, and every one of them would have silently altered a published result.

5.2 Before a number is quoted: adversarial audit

By adversarial audit we mean an independent recomputation of a claimed quantity from raw data by someone briefed to find it wrong. In this project the auditors were AI coding agents given that brief and, for the sealed test of Section 7, denied access to the answer key; the phrase does not mean the red-teaming of a model that it denotes elsewhere. An audit assumes the result is wrong, looks for the ways its authors have flattered themselves, and recomputes the quantity independently rather than re-running the pipeline that produced it, so that a defect in the pipeline cannot confirm itself. Independence of implementation is the point: on more than one occasion the recomputation agreed with the artefact and disagreed with the prose written about it.

The method’s value is measurable in what it caught. It found that a baseline was detecting file provenance rather than faults. It found that one of the fourteen single-duct detections was an artefact of the dataset (the fault-free file was simulated on a different configuration branch from every fault file, and the detector was noticing the branch, not the fault), so the score is reported as 14 of 14 naive and 13 of 14 adjudicated. It found that a day-level significance test adopted early in the work rested on an impossible null, the discordant cell it tested being zero by construction; the test was retracted, the significance it had awarded withdrawn, and a binomial against each channel’s own noise put in its place. It found detections below sensor precision, an evidence sentence falsified by its own recomputation, and, twice, a completed comparison against commercial practice: the first execution invalid for three independent reasons, the repaired one for six adapter defects and one defect of design (Section 6.4). None of these would have been caught by a test suite, because in each case the code was doing what it was written to do.

5.3 After a result: tests that pin numbers, falsifiers and caveats

Guards run at three levels. About two hundred automated tests execute on every commit through continuous integration; the data-dependent tests skip rather than error when the datasets are absent, so the suite is meaningful on a bare clone. A regeneration gate re-derives every committed artefact from source and diffs it against the stored copy, reporting per artefact either an identical verdict or a named set of changed values; a changed verdict is a defect or a documented decision, never a silent update. And every validated dataset is frozen permanently into the suite as a fixture, so the examination the method must pass only ever grows. The single-duct system is the byte-reproducibility standard: re-running its benchmark must reproduce the committed artefact byte for byte, and that requirement is enforced after every change.

The unusual property of the suite is what it guards. Alongside conventional unit tests of the event, residual, frequency and grammar logic, it pins three things not ordinarily thought of as testable. It pins published values, so that a change which would alter a number quoted in a paper fails the build rather than propagating silently. It pins fired falsifiers: the sampling-rate experiment’s finding is its four refutations, and a regeneration that loses them is a regression exactly as a lost feature would be. And it pins disclosure obligations: one test asserts that the artefact still declares which channels’ false-positive rows are calibration targets rather than out-of-sample measurements (the caveat reproduced in Section 8); another asserts that the disclosure that the rate-channel demotion was chosen after its effect was known ships alongside the result

it qualifies; a third recomputes the cry-wolf ratio from the benchmark and union artefacts rather than reading the stored value, so that an internally consistent but wrong number cannot survive; a fourth recomputes sensor coverage from the configuration files for the same reason. Continuous integration has been used to guard that a published result regenerates [37]. We have not found it reported that continuous integration guards that a published caveat survives. The justification is that caveats decay faster than code: a sentence in a document can be dropped in a rewrite without anything failing, whereas a test that enforces the sentence cannot.

5.4 What the protocol taught, and reproducibility

Sixty-seven numbered lessons record each failure with the standing rule it produced, and three patterns run through them. Lessons recur unless something enforces them: a rule requiring the set of days scored to be derived from the raw calendar rather than from the event log was recorded, documented, and violated again two phases later, so the standing policy is that a lesson is not closed until a helper or a test mechanically enforces it. Every new detection channel recreated the same gap: each needed its own noise-floor gate, and each time there was a temptation to exempt it, until the pipeline enforced the gate structurally. And simulation flatters: noise-free simulated years make zero-false-positive thresholds cheap, bands thin and silence easy, which is why the sensor-precision floor and the noise-floor significance requirement exist, why Section 8 treats the thinness of field data as the principal threat to these results, and why the one measured stream, the field rooftop unit of Section 6, is reported as a false-alarm rate and a case rather than as a detection result.

The datasets are public and citable by DOI [27–29]. The environment is pinned by lockfile, random seeds are fixed wherever randomness exists, and the repository documents the complete path from a fresh clone through data conversion, the data checks and the result pipeline to regenerated artefacts, with approximate wall-clock times. A claim-to-artefact table maps each published claim family (scorecards, joint false-alarm rates, baselines, the discovery-automation ablation, the grammar and reversal probe, the zone ground truth, the circularity bound, sensor coverage, cry-wolf ratios, the robustness experiments, the alarm attribution and the sealed transfer test) onto the specific artefact that backs it, so that a reader wishing to check a single number does not have to regenerate everything to find where it comes from. One provenance note illustrates the standard: the circularity bound of Section 4 compares the current pipeline against its pre-split predecessor, which lives in a different repository; rather than cite that repository, its artefact is committed here, so that both sides of the comparison regenerate from this one.

5.5 What the protocol found in the datasets

Auditing the evaluation produced an unanticipated secondary contribution: eight defects in the LBNL benchmark datasets, each with machine-verifiable evidence and a reproducing command (the creators’ own account of the models’ fidelity limits [14] is the closest prior; the defects here are byte-level and label-level findings, not fidelity findings) (Table 3). One of them, the configuration-branch mismatch (E5; row 4 of Table 9), hid for six phases because every comparison ran against the same fault-free file and no fault-versus-fault control existed.

The most consequential for other researchers concerns the supply-air static-pressure columns, which carry a per-branch additive constant of about 401.86 (added to one column in the fault-free file and subtracted from the other in every fault file) with no overlap between the two ranges. Either column alone separates fault-free from faulty data by file provenance rather than by fault physics: a single threshold on either column, chosen from the fault-free file alone, classifies all 365 fault-free days and all 7,150 days of the twenty fault files correctly (accuracy 1.0 on both

ID	System	Columns / files	Defect	Consequence for evaluation
E1	SDAHU	oa_bias (four severity files)	One simulation run shipped four times under different severity labels; the labelled sensor-side bias never appears in the logged column. The evidence is consistent with a bias injected only into the cooling interlock’s outdoor-air input, but observationally indistinguishable from a mislabelled cooling-lockout-setpoint fault.	The four severity labels index one run, not four independent scenarios, and the labelled sensor-side bias is absent from the logged column; the mechanism is not asserted.
E2	SDAHU	coi_leakage (four severity files)	One simulation run shipped four times under different severity labels (the scored count of 14 already collapses each such family to one scenario).	The four severity labels index one run, not four independent scenarios.
E3	SDAHU	SA_SP, SA_SPSPT	A constant of about 401.86 added to SA_SP in the fault-free file and subtracted from SA_SPSPT in every fault file (one column corrupted per branch, a different column each) with no overlap between the two ranges.	Either column alone separates healthy from fault data by file provenance rather than by fault physics; both are excluded from all baselines here.
E4	SFPU	SensorBias_ RMTEMP_-2C	The file’s calendar is rotated relative to its siblings.	Excluded here from family claims, so the SFPU scenario universe is scored without it.
E5	SDAHU	healthy file against every fault file	The healthy file was simulated on a different configuration branch than every fault file, with an occupied damper floor of 0.0 against 0.1 and different schedules.	Healthy-versus-fault divergence partly measures the branch rather than the fault; one single-duct scenario was detected through that provenance difference (14/14 naive, 13/14 adjudicated).
E6	FCU	Fouling_Cooling_ Airside_Minor, Fouling_Heating_ Airside_Minor	The two files are byte-identical: one simulation run shipped under two different family labels (a cooling coil and a heating coil).	One label is wrong and the data cannot say which; the content is scored once and the second entry is excluded, so the fan coil unit’s fouling family is 11 scenarios and its scored total 47.
E7	PFPU, SFPU	every fault file	The documentation places every seeded fault in the west zone (__W); in every file only the south zone’s columns (__S) move (57 of 60 files by the movement score, three indeterminate; the project’s earlier zone audit, a different instrument, gives 50 and 10).	A reader scoring localisation against the documented zone scores every correct alarm as wrong; the project’s zone ground truth was established from the data.
E8	DDAHU	SensorBias_ CSP/HSP_ -2inwg, -4inwg	The deck static column is unchanged; the four mixing-box statics shift by +0.20 and +0.40 in.wg, one tenth of the labelled bias (the +2°C temperature file shifts by exactly its label).	Severity ladders and per-in.wg sensitivity claims built from these labels are off by ten; detections are unaffected.

Table 3: Eight machine-verified defects in the LBNL benchmark datasets. Each has a reproducing command in the repository’s ERRATA.md.

columns), so any model given these columns can identify provenance perfectly without reference to any fault. We discovered this by auditing our own baseline, whose error statistic had reached implausible magnitudes for precisely this reason; the columns are now excluded from all baselines. Published results on this dataset that used the full column set should be read with this in mind. The companion paper [69] gives each defect with its reproducing command and the protocol that found it; here they stand as the evidence that the checks of this section catch what summary statistics do not.

6 Results

STRATA detects 158 of 175 seeded scenarios on the five simulated systems (14 of 14 naive on the single-duct unit, 13 adjudicated; totals throughout use the naive count, so 157 under the adjudication) at 26 false-alarm days in 480 held-out fault-free days, at a median of one day, and names the implicated terminal unit in 37 of the 50 attributable detections with none wrong (Tables 4 and 5). Every number in this section is scored by the detector’s final calibration, the per-day noise floor of Section 4; where a result was measured under an earlier gate, the text says so. What follows reports how the detections are made, which channels carry them, what the process-mining channels contribute, and what remains missed.

6.1 Detection and false alarms

Evaluation began with three public LBNL fault-detection datasets [28, 29]: a single-duct air-handling unit (SDAHU) and parallel and series fan-powered-unit configurations (PFPU, SFPU); four further datasets were onboarded afterwards (the dual-duct and fan coil units in Section 6.9; the two rooftop units in Sections 5 and 7), and Table 5 lists all seven datasets with Figures 2 and 3 summarising detections and false alarms on the six scored ones. The two fan-powered-unit datasets had not previously been benchmarked in the published literature: the searches described in Section 3 returned no published fault-detection result on either. The fan coil archive, by contrast, has since been used by an autoencoder study [46] and by a benchmark-validity audit [65]; no published detection result on the dual-duct unit was found. Scoring follows the day-level, scenario-as-unit conventions of the FDD evaluation protocol literature [22, 89]. Four counts recur in what follows and each includes a different set: seven datasets went through the pipeline; six are scored for detection (the five simulated annual systems and the simulated rooftop unit; the measured rooftop unit has one fault case and contributes a false-alarm rate only); the five simulated annual systems carry the headline totals, because their holdouts are 96 days each and pool to 480; and the three original systems (single-duct, parallel and series) carry the two baseline comparisons, which were run on their 73 shared scenarios.

6.2 Time to detection, and how to read recall

The median time to detection is one day on all six systems, but the median alone misrepresents the distribution and should not be quoted without its tail (Table 6).

Gradually developing faults are detected slowly (the two 114-day tails on the dual-duct unit are its moderate and severe waterside cooling-fouling files), which is the expected cost of calibrating below a noise floor and is a genuine limitation for the degradation faults that motivate much of this literature.

Sensor-bias detection illustrates why pooled recall misleads for gated detectors. Raw recall of 39–47% over the 351 to 352 evaluable days appears to be a capability ceiling until it is de-

	SDAHU	PFPU	SFPU	DDAHU	FCU
Scenarios detected	14/14 naive; 13/14 adj.	26 / 30	25 / 29	50 / 55	43 / 47
Median time to detection	1 day	1 day	1 day	1 day	1 day
Deployed false-alarm rate	1.0%	8.3%	9.4%	3.1%	5.2%
Naive all-channel union	15.6%	15.6%	9.4%	8.3%	21.9%
Wrong alarms / all alarms	0.03%	0.11%	0.12%	0.02%	0.04%

Table 4: Detection performance on the five simulated systems over the 96-day healthy holdout (the last eight days of each month). Naive and adjudicated refer to before and after the branch adjudication of Section 7; the cry-wolf ratio is wrong alarms as a share of all alarms ($\leq 0.13\%$ on every system, from 96 fault-free holdout days against 4,540–16,546 observed fault-scenario days), and is small partly because observed fault-scenario days outnumber fault-free days 47–172 to 1. The SDAHU score is reported in both forms for the reason given in Section 7; the SFPU universe excludes one file with a rotated calendar (erratum E4); the two false-alarm rows differ by the demotion of the seasonal rate channel disclosed in Section 8, and coincide on the series unit, where that channel contributes no false-alarm day. Since the three-way split every deployed rate is out-of-sample (Section 5).

System (LBNL)	points	rules (state / sig.)	scored	detected	false alarms	healthy days
SDAHU, single-duct AHU	26	3 / 12	14	14 (13 adj.)	1 / 96	365
PFPU, parallel fan-powered box	109	13 / 45	30	26	8 / 96	365
SFPU, series fan-powered box	109	13 / 49	29	25	9 / 96	365
DDAHU, dual-duct AHU	114	15 / 56	55	50	3 / 96	365
FCU, fan coil unit	29	7 / 19	47	43	5 / 96	365
RTU, simulated rooftop unit	23 [†]	3 / 5	24	20	0 / 28	100
RTU, measured rooftop unit (Site 2)	26 [†]	2 / 7	1 case	—	3 / 49	182

Table 5: The seven LBNL datasets as scored by the final calibration: what each configuration maps and declares after the coverage audit of Section 6.10, what was scored, and the out-of-sample false-alarm count on the held-out fault-free days (the last eight of each month). The measured unit carries one fault case and is used for its false-alarm rate only. [†]Includes one derived operating-state column (fan-on for the measured unit; a constant for the simulated unit, which has no schedule).

System	Detected	At 1 day	At 2 days	Remaining TTDs (days)
SDAHU	13 (adjudicated)	9	4	0
PFPU	26	26	0	—
SFPU	25	25	0	—
DDAHU	50	45	0	3, 5, 14, 114, 114
FCU	43	38	1	3, 3, 4, 15
RTU (simulated)	20	19	0	14

Table 6: Time-to-detection distribution behind the one-day median, by system, as scored by the final calibration (from the committed scorecards). The long tails are fouling and charge faults: on the dual-duct unit the moderate and severe waterside cooling fouling (114 days each, carried by the rules channel), on the fan coil unit the moderate heating waterside fouling (15 days, the operating-point indicator of Section 6.10), and on the simulated rooftop unit the 20% refrigerant undercharge (14 days).

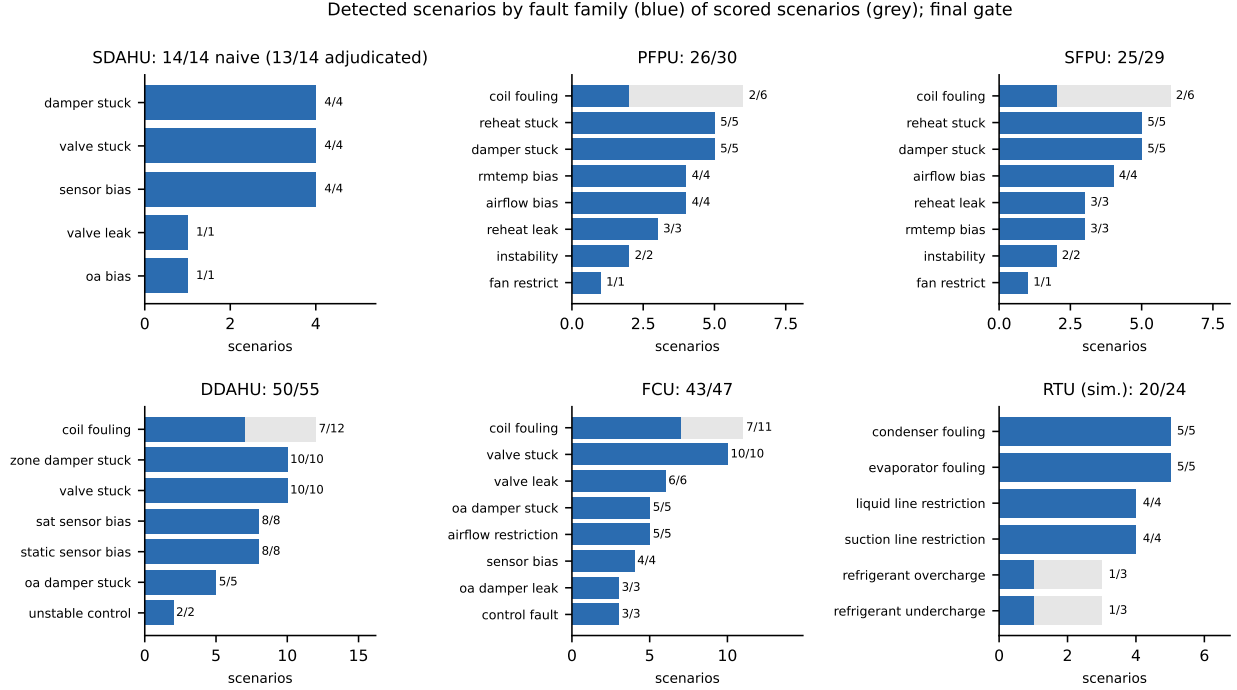


Figure 2: Detected scenarios (blue) of scored scenarios (grey) by fault family on the six scored systems, as scored by the final calibration. Coil fouling is the dominant miss on the four air-side datasets and is caught in full on the simulated rooftop unit, condenser fouling through the refrigerant side and evaporator fouling through the air side (Section 7).

composed. The residual is physics-bound only inside a detection window (cooling coil off, zone occupied, sustained at least 120 minutes) and such windows exist on 137 to 164 days of the simulated year. Within them, window-conditional recall is measured at 100%, on counts of 164/164, 164/164, 139/139 and 137/137 across the four coil-bias scenarios. The apparent ceiling is window prevalence, not capability, and reporting that prevalence separately from window-conditional recall is a methodological recommendation we would make to others evaluating gated detectors.

6.3 Against a principal-component baseline

Against a principal-component-analysis baseline using squared prediction error [35, 84], audited and scored on comparable scenario universes, STRATA detects three more scenarios than the baseline inside overlapping intervals (Table 7). The baseline standardises every raw daily feature, fits principal components to 95 % of the variance of the fault-free training days, scores each day by its squared prediction error, and takes as its threshold the largest training-day score: the strict form quoted here, which is the same extreme calibration STRATA’s bands use; a 99th-percentile variant is committed beside it. The provenance-leaking static-pressure columns of Section 5.5 are excluded from its features.

The difference that matters operationally is not the count. The baseline does not name the implicated device or state the evidence behind a flag; Figure 4 shows what STRATA reports on one seeded fault, and it is the property a scalar score cannot produce. On the count, the misses are complementary rather than nested: three scenarios are detected only by STRATA and none only by the baseline. STRATA alone detects an unstable VAV damper and the parallel unit’s moderate

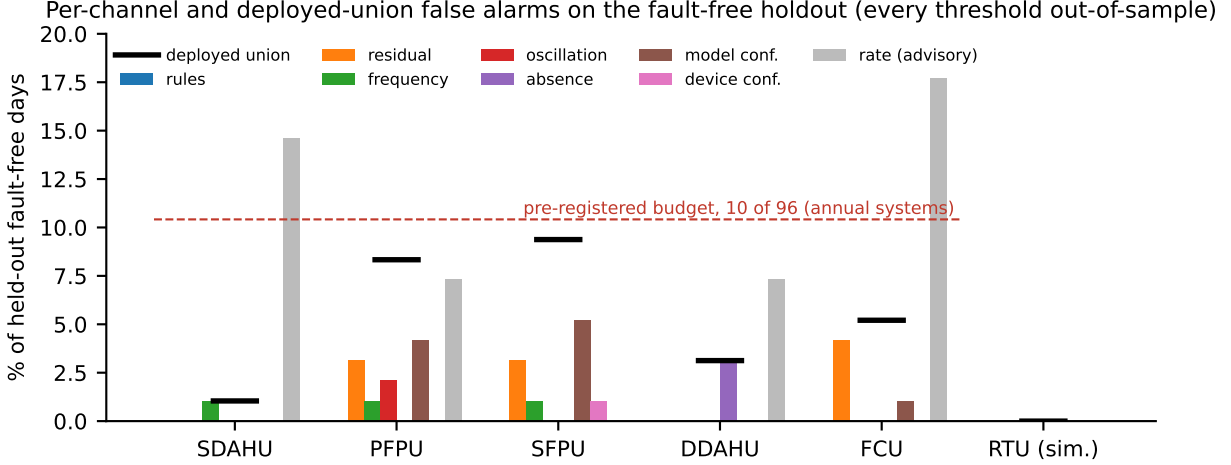


Figure 3: False alarms on the held-out fault-free days, per channel and for the deployed detector as a whole, every threshold out-of-sample. The seasonal rate channel (hatched) is advisory and outside the deployed union; the dashed line is the pre-registered budget of 10 of 96 held-out days for the five annual systems; the simulated rooftop unit’s 28-day holdout carries no such budget line.

and severe airside reheat fouling. Before the coverage audit of Section 6.10 the split was one against six, the baseline reaching three room-temperature biases, two series-unit airside fouling files and the series unit’s fan-flow restriction through zone-fan columns that STRATA’s configuration did not then map; every one of those six is now detected. Nine scenarios are missed by both. Eight of them are one family: reheat-coil fouling, waterside at every severity on both units and airside minor on both, is invisible to both methods; the ninth is the single-duct outdoor-air-bias run, a miss only under the branch adjudication of Section 5.5. That is the most useful finding in the comparison: reheat-coil fouling, at the severities seeded here, is invisible in the recorded signals to a learned detector and to a rule-based one alike, which is information for anyone choosing what a terminal unit should measure. On false alarms the baseline is better on the two terminal-unit systems and within one day of STRATA on the air handler, on a different day universe; the intervals in Table 7 overlap throughout. On detection, 64 of 73 is [78.2, 93.4]% and 61 of 73 is [73.4, 90.3]% (Wilson), overlapping; an exact McNemar test [49] on the three discordant scenarios gives $p = 0.25$, which with three pairs is an absence of power rather than evidence of superiority.² Nor are the inputs symmetric: the baseline consumes 108 and 424 raw daily features on the single-duct and fan-powered systems where STRATA’s configurations mapped 12 of 30 and 56 of 109 columns when the comparison was first run and map 26 and 109 since the coverage audit (Table 7), so the inputs are now comparable in breadth but not in kind: the baseline consumes raw columns, STRATA physically named relations between them.

6.4 Against current practice: the Guideline 36 rule battery

The comparison that matters most to practitioners is against the rule-based detection actually deployed in commercial building automation systems [30]. The result first. Held to the treatment STRATA gives its own signature rules, a correctly wired third-party Guideline 36 battery detects

²Every interval and the paired test regenerate from the committed artefacts: `scripts/intervals.py` $\rightarrow$ `outputs/intervals.json`, under the regeneration gate.

	STRATA	PCA-SPE (strict)
Detected (of 73, adjudicated)	64	61
Per system	13 / 26 / 25	13 / 23 / 25
Holdout false-alarm rate	1.0% [0.2, 5.7] / 8.3% [4.3, 15.6] / 9.4% [5.0, 16.9]	1.3% [0.2, 6.8] / 1.0% [0.2, 5.7] / 1.0% [0.2, 5.7]
Sensor columns mapped (STRATA)	26 of 30 and 109 of 109 (12 and 56 before the coverage audit)	—
Raw daily features consumed (PCA)	—	108 and 424
Implicated device named correctly	37 of 50 (13 unnamed, 0 wrong, 1 indeterminate; the 14 single-duct detections have no device stratum; naive universe)	not applicable
Names the implicated device	yes	no

Table 7: STRATA against the principal-component baseline. Per-system entries are ordered SDAHU / PFPU / SFPU; the counts behind the false-alarm rates are 1/96, 8/96 and 9/96 for STRATA and 1/79, 1/96 and 1/96 for the baseline, with Wilson 95% intervals [86] in brackets. Feature counts are given for the single-duct and the fan-powered systems.

12 of 14 on the single-duct air handler at one false-alarm day, against STRATA’s 13 at one; and 8 of 30 and 9 of 29 on the fan-powered terminal units at no false-alarm day, against STRATA’s 26 and 25 at eight and nine. STRATA is within one detection and one false-alarm day of current practice on the air handler, and detects three times as many seeded faults on the terminal units at the price of eight or nine false-alarm days in 96; the difference on the terminal units is coverage, the per-zone residual readers of Section 6.10, not the gate. The pre-registered refutation condition, that the battery matching STRATA’s count at an equal or lower false-alarm rate on any system would end the improvement claim there, fired on every system, and the claim is withdrawn.

The design imports a third-party open-source implementation of the Guideline 36 fault conditions [4], open-fdd 4.4.1 [6], released 2026-08-15 and pinned in the repository, rather than reimplementing them, because a home-made version of a competing method invites the objection that the comparison was won by construction. Parameters stay at the library’s installed defaults; air handlers are graded with the fifteen air-handler conditions FC1 to FC15 and terminal units with the library’s terminal-unit rules, run per zone; both methods sit the identical examination. Two executions preceded the one reported. The first fed two coil conditions water-side temperatures where the standard specifies air-side ones, graded terminal-unit faults with the air-handler rulebook, and read the provenance-leaking static-pressure columns of Section 5.5; it was voided. The second, under a pre-registered repair, held every prediction, with the battery flagging 77, 96 and 96 of 96 fault-free days and detecting nothing under the gate, and was then voided by two independent audits that found six wiring defects in the repair (a fan-speed column that is a constant in the dataset, a terminal-unit gate on the box fan of a parallel unit, a hunting rule reading a binary fan status, night-cycle graded as occupied, a single-duct leaving-air rule applied to boxes that mix plenum air, an outdoor-air condition reading a column not in cfm) and one defect of design: a gate applied once to the pooled union of all rules, which can pass nothing once the union flags every healthy day. The amendment recording all of this was committed before the third execution, whose

STRATA alarm -- series fan-powered unit, 2018-01-01 (day 1 of the fault)
 what zone primary-air flow not tracking its setpoint [channel: rules]
 where terminal unit TU_S, zone S
 evidence measured 715 CFM against a 200 CFM setpoint -- a 515 CFM excess
 sustained for 720 min while occupied (rule fires at 50 CFM held
 60 min); damper position 0.80, fault-free same-day value 0.32
 budget rules channel: 0 false-alarm days on the 96-day fault-free
 holdout; threshold set on training days only, holdout out-of-sample
 corrob. seasonal rate channel agrees (advisory; never alarms alone)
 seeded zone-S damper stuck at 80% -- location correct, symptom correct,
 detected on day 1 of 365

Figure 4: A worked alarm. Scenario SFPV_VAVDMPRStuck_80%; rule zone_flow_tracking_S, a setpoint-deviation rule gated on occupancy. On day 3 the unit-model conformance channel also flags; this is the scenario on which a one-line rule later out-covered that model (Section 7). The alarm names the symptom and its location, not the root cause; a later pre-registered layer that maps signed residual patterns to fault families names the family in three of four resolved cases and is not yet adoptable (Section 6).

artefacts are the ones quoted.³

Three framings were then pre-registered and the falsifier applied to each. Gated once on the pooled union, the battery detects 1, 0 and 0 scenarios, because the union false-alarms on 49, 96 and 96 of 96 held-out days. Gated per rule against each rule's own held-out false alarms, STRATA's per-channel structure, it detects 12 of 14 on the single-duct unit and every scenario on both fan-powered units; but on those units the three conditions that carry the verdict flag the same days on the fault-free file as on every fault file, and pass only because each rule fires less often in the held-out days than across the year, a drift the held-out floor cannot see and that STRATA's channels never present because onboarding silences every rule on the whole fault-free year first. That framing is reported as a property of the floor, not as detection; the lesson is that a held-out null is valid only for a channel whose rate is steady across the year, which is what silence on the fault-free year buys. The third framing treats the battery exactly as STRATA treats its own signature rules, keeping the rules that are silent on the fault-free year and holding them to the same 3-in-365 null [32]; it gives the counts stated above. The two single-duct misses are the 10 % and 25 % stuck dampers, seen only by a condition that is not silent; the terminal-unit detections are the unstable room loop, the larger stuck and leaking reheat valves and the stuck dampers.

A re-audit found no wiring defect and three things a reader must weigh. Without the rules that detect the fault-free file itself, the per-rule counts on the fan-powered units are 15 of 30 and 17 of 29. The silent-rule selection is made on the whole fault-free year including the held-out days, on both sides, so the single-duct tie of one false-alarm day each is like for like. And the single-duct outdoor-air-bias file, adjudicated away on STRATA's side (Section 5.5), is counted on the battery's, where its carrying condition passes the same control; under every reading STRATA detects more on that unit, and the match rests on the false-alarm tie alone. Three caveats travel with the numbers. The parameters are open-fdd's installed defaults, which are tighter on every condition that fired than the Guideline 36 values the library's own labels cite (sensor errors 1.15 °F against 2 to 5, fan heat rise 0.55 against 2 °F, no mode-change delay against 30 minutes); with the standard's values the fan-powered air handler's 2.15 °F fan heat rise would largely not trip the condition (FC8) that

³docs/plans/2026-09-27-x38-guideline36-repaired-prereg.md and its Amendment 1; outputs/x38_guideline36.json; earlier executions kept under outputs/void/ and as *_run1.json.

carries the battery’s one single-duct detection STRATA adjudicates away. The terminal rules carry fixed comfort bands and flow thresholds that do not read the building’s own setpoint columns. And supplying the one site value every deployment supplies, the design minimum outdoor-air flow, left the outdoor-air-fraction condition alarming on 94 of 96 healthy days, because the implementation has no operating-state gate.

6.5 What the discovered models contribute: matched rules and ablation

The matched-rule control isolates what discovery contributes. For each channel a hand-written equivalent was authored and scored. Two of these reproduce their channel exactly (on the unstable-room-loop scenario, 141 flagged days against the frequency channel’s 141 on the series unit and 206 against 206 on the parallel unit), which is the point: the stratified log’s own count bands make explicit rules an expert could have written, had they known where to look (the matches are against a hand-written per-zone heating-episode band, located by the strata, not by the discovered net). Transplanted without that knowledge the same rules are unusable, one of them firing on 231 fault-free days on the single-duct system and 124 on the parallel unit while firing on none at all on the series unit: but the single-duct figure is not evidence about the models: that system has no heating rule and no hot-water valve, so a rule reading ‘heating never active’ fires on every occupied day for want of a sensor, not because any model predicted it. The only informative contrast is the parallel unit against the series unit, and whether the discovered models predict that contrast is the subject of a pre-registered test reported in Section 7. Some faults, conversely, need the classical instrument: a calibrated oscillation channel stands alone on the series unit’s unstable-damper scenario, 349 flagged days against at most two from any matched rule (264 on the parallel unit’s).

The decisive ablation is the one the matched-rule control implies. Reading the significance-gated channel credited with each detection from the committed scorecards, no scenario on any system is detected only by the two conformance channels: they are credited with none of the 14 naive (13 adjudicated) single-duct or 26 parallel-unit detections, and appear in 7 of the 25 series-unit detections but never alone (counts after the coverage audit of Section 6.10, which left the model and device rows identical). Removing them leaves every scorecard unchanged. On the series unit they appear on 6 of the 9 deployed false-alarm days (out-of-sample since the three-way split); removing them eliminates five (the sixth is also flagged by another channel), lowering that system’s rate from 9/96 to 4/96 at no detection cost, and takes the parallel unit from 8/96 to 6/96 and the fan coil unit from 5/96 to 4/96 (recomputed from the per-channel day lists in the union artefacts). The discovered models contribute the strata and the invariants that shaped the rules; they do not contribute detections.

6.6 Localisation and portability

Every alarm carries a location, and the location is right wherever it can be checked. On the 51 detections on the two fan-powered systems in the attribution universe (65 naive detections less the 14 on the single-duct system, which has no device stratum; one with indeterminate ground truth), the alarm names the correct terminal unit in 37, names none in 13, and names a wrong unit in none, three wrong units were available in every case, so the zero is falsifiable even though the 37 is not. Localisation is reported as specificity rather than discrimination: because every seeded zone fault in the benchmark occurs in the same zone, the ground truth has no variance and a perfect score would be unfalsifiable as evidence of discriminating between zones, so no per-zone discrimination figure is quoted. A second, pre-registered attribution test asks where in the equipment the alarm points: over all 158 detections on the five simulated systems, the rule carrying

the detection names the seeded component exactly in 109 (69 %) and the right subsystem but a neighbouring component in 22 (14 %), which is 83 % at subsystem level, the wrong subsystem in 20 (12.7 %), and no component in 7 (four carried by the frequency or oscillation channel alone, three by residual rules the attribution table does not name). The pre-registered bar for a component-level claim (wrong subsystem ≤ 10 %) was missed, so the claim is subsystem-level. Fourteen of the twenty wrong cases are on the single-zone fan coil unit and name the component that responds to the fault rather than the one that caused it: the coil for a biased room sensor, the fan for a fouled coil, the valve for a reversed controller (the first place a technician would look, but not the cause.⁴ A pre-registered diagnosis layer built from three fault-agnostic ingredients) the sign of every significant residual against its healthy band, a consensus rule over redundant copies of one quantity (the four mixing-box statics of a deck, the four return-minus-zone residuals), and a leave-one-out nearest-pattern vote for the fault family, raised the exact-component count to 112 of 158 (70.9 %), 85 % at subsystem level, and lowered the wrong-subsystem rate to 10.1 %: the consensus rule resolved the five dual-duct damper cases the carrying rule had scored wrong (one box copy moves alone) and called 14 of 16 deck sensor biases at the deck (all four copies move the same way), which is the mechanism the pre-registration described, working where the redundancy is real. The layer did not clear its bars: the family is named from the signed pattern alone in 85 of 114 resolved detections (74.6 %, bar 80) with 44 unresolved (27.8 %, bar 15), and the wrong-subsystem rate of 10.1 % missed the bar (below 10) by one case. It does nothing on the fan-powered units, where a single zone’s fault moves that zone’s return-minus-zone residual one way and the other three the opposite way because the return is the mix; and the family confusions that remain (a stuck damper against a biased airflow sensor, a restricted filter against a fouled coil, a leaking against a stuck valve) are pairs the recorded quantities cannot separate by sign.⁵

Portability was measured. The second building type was onboarded through configuration alone in approximately 1.5 hours of wall-clock time, reconstructed from commit timestamps, comprising 45 sensor mappings and 35 rules at onboarding (109 mappings and 58 rules in the committed configuration, after the coverage audit of Section 6.10); the third was derived from the second in minutes. No pipeline code changed.

6.7 What the event log carries, and the time perspective

The null of Section 6 was argued from the model side (order-permissive nets, count deviations surfacing as alignment cost) until the log itself was examined. For the parallel fan-powered unit we compared, day by day, the fault-free year against a missed scenario and a detected one on the state alphabet alone. Under waterside reheat-coil fouling at its most severe the state-event log is identical to healthy to the minute: 162.8 against 162.4 events per day, the same activity counts, a zone-S heating-on median of 124 minutes in both. Over all 73 scenarios, at a 5% tolerance on every activity’s daily count and every state’s on-duration median, eight of the twelve scenarios the deployed detector missed at the time have a state log indistinguishable from healthy (all reheat-coil fouling; nine files in all have that property, and every one of the eight fouling misses that remain on these two units under the final calibration is among them) and for them no method that reads that log can detect, process mining or otherwise, because at the deployed alphabet the fault lives in heat-transfer magnitudes the abstraction discards; the other four change a daily count by 7–39% that the frequency channel’s bands did not resolve into significant days.⁶ Under a 50% reheat-valve

⁴outputs/component_attribution.json; docs/plans/2026-09-25-x34-component-attribution-prereg.md.

⁵outputs/x37_diagnosis.json; docs/plans/2026-09-27-x37-diagnosis-layer-prereg.md; three runs, two script defects corrected to the pre-registered wording, earlier runs kept.

⁶outputs/x12_log_diagnosis.json.

leak the log does change (the zone-S heating-on median falls from 124 to 11 minutes and the zone fan cycles 26.6 times a day against 16.3) in timing and in counts, with the activity set and its order unchanged. Control-flow conformance is blind to timing by construction; the frequency channel already reads counts.

That diagnosis fixed a pre-registered test of the one perspective never tried (experiment X12, in the numbering of Appendix A): a time-infused state model: for every on/off pair of the state alphabet, the day’s median on-duration, off-gap and first-on minute, train min/max bands widened by five minutes, the holdout false-alarm floor and the same binomial gate as every channel. Against the committed scorecards’ flag days it is significant on 11 of 73 scenarios with p down to 10^{-94} (10 under the more conservative rule-of-three floor), on none of the twelve fouling scenarios (predicted), and on no scenario the deployed channels miss (the load-bearing prediction, refuted); it is never earlier than the deployed detector (typically day 14 against day 1) and it costs five holdout false-alarm days on each fan-powered system, taking the parallel unit’s then-deployed joint rate from 5.2% to 10.4%, over budget. What it adds is a small day-level coverage supplement on scenarios already caught, 81 unique days on the single-duct coil-bias run (15 against the gated union), 16 on a series-unit sensor-bias run, none on five further series-unit scenarios, four reheat-valve faults and a stuck VAV damper, that the rules already flag on every day, through physically legible statistics such as zone-heating on/off medians and the cooling first-on minute.⁷

A fault-informed coil-effectiveness residual for the fouling misses (X13, design disclosed as chosen after inspecting three fouling files, thresholds from healthy train days only) then fired its falsifier: one of twelve fouling scenarios significant, nothing newly detected. A last pre-registered variant asked whether the model had simply been starved of vocabulary (X14): healthy-derived band states for every actuator position and flow (above the occupied 90th percentile, at or below the 10th, fifteen-minute dwell; the wall intact) enrich the alphabet by 6, 40 and 36 events. The single-duct model channel then fires on all 14 scenarios instead of none, and one of the twelve misses, moderate airside fouling of a series-unit reheat coil, becomes significant on the frequency and time channels, at a holdout false-alarm rate of 7, 18 and 21 days in 96 (on the calendar-day universe), over the budget on both fan-powered units, where alignment-based conformance on the enriched log was also computationally impractical (single-laptop timings, printed and not committed: seventeen minutes to fit; one scenario not scored in forty-seven) and was not evaluated; X14’s falsifier fired.⁸ The detection is carried by one added state, the zone damper’s low band, entered nine times a day against under one on the healthy year; a committed fault-versus-fault control over all 31 series-unit files shows a monotone ladder (64%, 64%, 12% and 3.5% of occupied samples at or below the band edge from healthy to severe), with severe firing on the same statistic and already detected, and the three waterside-fouling files sitting exactly at the healthy values at every severity, so it is a physically legible, dose-monotone signature of airside fouling that the deployed alphabet did not carry, on channels that false-alarm one clean day in five.

A post-hoc read of the same artefact, not a pre-registered result, locates the cost in the time model (15 and 19 false-alarm days in 96) and finds the frequency channel alone, on the enriched alphabet, at 4 and 3 in 96 and still carrying the detection. That read was then pre-registered as a test with its provenance stated and the holdout as the thing that could still falsify it (X15): the enriched frequency channel alone stays inside the budget under both the deployed and the mirrored eight-day holdout (4 and 8, 3 and 5 days in 96 on the fan-powered units), detects the airside-moderate fouling scenario under both, adds nothing else new, a scenario the principal-component

⁷outputs/x12_time_perspective.json; pre-registration and fourth prediction’s weakness disclosed in PHASE9_RESULTS.md.

⁸outputs/x14_enriched_alphabet.json; three amendments, each recorded before the result it governs, in the pre-registration.

baseline also detects from raw features (Section 6), so the gain is within reach of a simpler method and the claim is only that an enriched event log carries it too, and is significant on 41 of the 61 scenarios the naive scorecard already detects against the deployed frequency channel’s 12, at three to eight times that channel’s own false alarms (1 in 96 on every system), which is the like-for-like price of the coverage. No falsifier fired. It is the one constructive result of the four vocabulary experiments (X12–X15): a candidate deployable addition, not an adopted one, since adoption changes every published scorecard and is a separate decision, to be pre-registered.⁹ The reading is now measured rather than argued: process mining’s contribution on these systems is the event log and its count and time perspectives; the discovered net’s order structure contributes nothing at the deployed vocabulary, and richer vocabulary buys detections at false alarms the budget forbids.

6.8 What the conformance channel can see: a positive control

The null above can be read two ways: the faults do not change event order, or the channel cannot see order. A pre-registered positive control (Figure 5) injected order violations (whole-day reversal, adjacent swaps, deletions, a deleted start event, a duplicated block) into each system’s fault-free holdout traces and scored them with the deployed net, threshold and significance gate, adding two threshold-free measures: the AUC of perturbed against unperturbed per-day fitness, and the hit rate of a threshold set at the fault-free holdout’s worst day. The channel turned out to be three instruments. On the fan coil and dual-duct units a reversed day is flagged on 55 of 72 and 59 of 83 holdout days (AUC 0.99; 75 and 71 % hit rate at zero false alarms): the channel sees order, and those units’ faults do not change it (the dual-duct unit joined this group only once its threshold was calibrated out-of-sample, Section 8). On the single-duct unit a reversed day is separable on average (AUC 0.91) but the hit rate at zero false alarms is exactly zero: the fault-free holdout contains days that fit worse than a reversed day, so no threshold inside the budget can see order: blind by the spread of healthy fitness, not by the net. On the two fan-powered units fitness does not move under any perturbation (AUC 0.36–0.51; deleting the start event raises it): blind by net structure. Eight adjacent swaps and four deletions are also significant on the two order-seeing units, consistent with alignment cost behaving as a graded detector there. No falsifier fired, and four of the five predictions were wrong.¹⁰

The instrument was then varied. A pre-registered grid of seven process-mining instruments (the inductive miner at noise 0.0 and 0.5 with alignment fitness, the deployed net with token-replay fitness, the heuristics miner with alignments, the log skeleton, Declare constraints and the temporal profile) was discovered on each system’s training days, thresholded at the 1% quantile of the per-day score on the calibration slice and scored through the deployed significance gate on the fault-free holdout and on the 33 scenarios the deployed detector then missed (8, 8, 10 and 7 on the fan-powered, dual-duct and fan coil units; sixteen of them have since been recovered by the coverage audit of Section 6.10, through columns no process model was ever given). No cell detects any of them at its threshold (realised false-alarm rates 0 to 9.7%; no threshold sweep was run); the highest share of scenario days any cell flags is 5.6%, by the log skeleton on a severe water-side fouling file, at 7 false-alarm days in 72. No cell exceeds 10% false alarms, so the alternative instruments are silent rather than noisy. Two heuristics-miner cells exceeded a 15-minute budget and are reported as not evaluated, not as nulls. The conformance null is therefore a property of the state-event log, not of the miner or the measure: what the log does not carry, no conformance instrument over it

⁹outputs/x15_enriched_frequency.json; the control in outputs/x14_control.json; per-channel false-alarm attribution in outputs/x14_enriched_alphabet.json.

¹⁰outputs/x22_positive_control.json; docs/plans/2026-09-24-x22-conformance-positive-control-prereg.md.

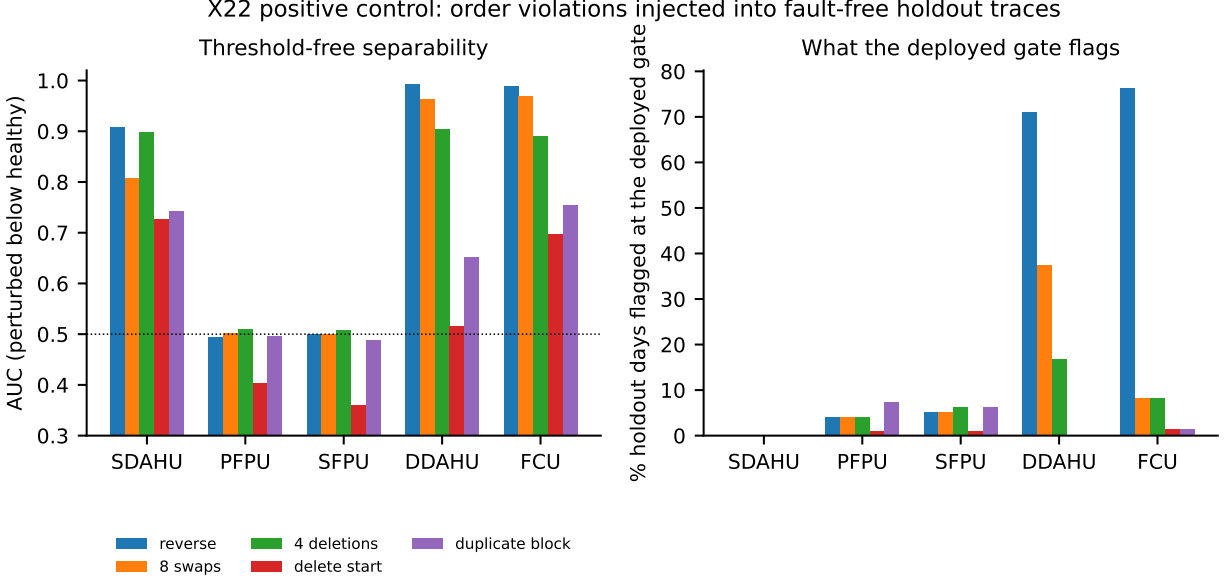


Figure 5: The positive control. Left: threshold-free separability (AUC) of perturbed against unperturbed fault-free holdout days for each injected order violation. Right: the share of those days the deployed gate flags. The fan coil and dual-duct units see order; the single-duct unit separates it on average but not at any in-budget threshold; the fan-powered units’ nets accept any order.

recovers.¹¹

6.9 A fourth and a fifth system, onboarded blind

The onboarding claim (that a building is a configuration directory, not code) was finally tested on a system type the project had never seen, under a pre-registration written after the documentation and before any data file was opened: the LBNL dual-duct AHU (two decks, two coils, two supply fans, three dampers, four zone mixing boxes with command-only dampers; 114 points; 55 seeded faults in 15 families, two of them (static-pressure sensor bias and unstable control sequences) absent from the first three systems). The configuration took 80 canonical sensor mappings (79 recorded columns, as Table 8 counts them, plus the timestamp), 15 state rules and 26 signature rules at onboarding (56 in the committed configuration, after the residuals added by later experiments and the coverage audit) reusing the existing rule kinds, and one healthy-silence iteration; no source change. That iteration is the instructive part: ten healthy days on which the cold-deck fan saturates and cannot hold its static setpoint drove nearly every healthy firing, and they were removed by a physical gate (duct static at setpoint) rather than a looser threshold. The gate battery, extended with a configuration-branch comparison of the kind that found E5, found no defect in the dataset at onboarding; the static-bias label defect E8 was found later, by a hostile review of this unit’s coverage audit.

Under the final calibration before the coverage audit, the detector caught 45 of 55 scenarios at 3 false-alarm days in 96 (8 with the advisory rate channel counted; 50 of 55 after the coverage audit of Section 6.10, which brings the static-pressure and deck-temperature biases to eight of eight each and the coil fouling to seven of twelve): every zone-damper, outdoor-air-damper, coil-valve

¹¹outputs/x30_method_grid.json; docs/plans/2026-09-24-x30-method-grid-prereg.md.

and unstable-control scenario, seven of eight deck-temperature biases, five of twelve coil-fouling scenarios (the pre-registration predicted at most four) and six of eight static-pressure biases (the sixth through the fan-law static residual added later). The discovered model channel is credited on two cold-deck damper scenarios that four other channels already carry (none before the three-way split); nothing was detected by the conformance channels alone. The enriched-alphabet frequency channel of X15, re-run here under its own protocol, stays inside the budget and triples its coverage of already-detected scenarios but catches none of the ten misses: the one fouling gain of X15 was a one-system signature and does not replicate. One check misfired usefully: the false-alarm script reported that demoting the rate channel would delay one scenario’s first alarm, and recomputation showed the day-one alarm belonged to the absence channel, whose day list the scorecards do not export, a blind spot in the check, now recorded as such, and no cost of demotion. Two things follow upstream. The transfer test’s scorable pool grows from two buildings to three (its amendment excluded the single-duct system structurally, for want of a heating activity), so the building-level floor moves from $1/2! = 0.5$ to $1/3! = 0.167$ with the dual-duct unit and to $1/4! = 0.042$ with the fan coil unit below, the first design that can attain $p < 0.05$; that re-run is row 10 of Table 9. And the dataset defects are properties of particular archives (every one of the five carries at least one, four of them on the single-duct unit), not of the collection.¹²

The fifth system, the LBNL fan coil unit (a four-pipe single-zone unit with a three-speed fan, both coils and one outdoor-air damper; 29 points; 48 seeded faults in 17 families, four of them (filter restriction, outdoor-air inlet blockage, fan outlet blockage and reverse-acting control, scored as two) absent from the first four), was onboarded under the same pre-registration form: 29 mappings, 7 state and 16 signature rules at onboarding (17 after the outdoor-air-fraction residual described below), two logged healthy-silence iterations, no change under src/ (the gate script needed two lines to read this system’s Brick file). The hash gate found a byte-identical pair under two family labels (cooling and heating airside fouling, minor); the content is scored once and the fouling family is 11, the scored total 47. The first run detected 47 of 47 at 28 false-alarm days in 96 and fired two falsifiers. Twenty-five of the 28 were the model channel’s, every one a Saturday or Sunday (the other three were residual days, and they survive into run 2): the two scoring scripts flag an event-less scheduled day as a model violation and define scheduled as any non-zero mode (a rule written for the fan-powered units, whose mode 2 is a night cycle that generates events) and on this unit mode 2 is idle setback, so 97 of 104 weekend days were “silent while scheduled.” The scorecard’s model gate then judged six scenarios significant against a conformance-only baseline (1 of 72 event days) that the deployed count (25 of 96) did not use; their 100 model days each were the healthy year’s own 100 weekend model days: 97 silent and three two-event traces. A two-script repair (scheduled means operate mode, the definition the occupancy event kind already used, and the gate’s baseline counts what the deployed channel counts) was pre-registered with four predictions and three falsifiers before it was made. All four held: the four earlier scorecards regenerated byte-identically, the fan coil unit scores 41 of 47 at 4 false-alarm days in 96 (residual 3, model 1), exactly the six model-only scenarios dropped and nothing else changed status, and nothing is detected by conformance alone (a pre-registered outdoor-air-fraction residual, Guideline 36 FC6 lineage, later caught the 20 % damper leak at unchanged false alarms, and the per-day residual null of the statistical repair, Section 8, took it back again: 40 of 47 under the final calibration); the model channel is credited on six scenarios under the final calibration, each also caught by rules, residual, frequency or oscillation. Both new families are caught in full (airflow restriction 5 of 5, control 3 of 3); the misses at onboarding were five of six waterside-fouling files (four after the coverage audit of

¹²outputs/x17_onboarding.json, benchmark_v6_ddahu.json, union_fpr_ddahu.json, week0_audit_ddahu.json; pre-registration and one amendment in docs/plans/2026-09-24-x17-ddahu-onboarding-prereg.md.

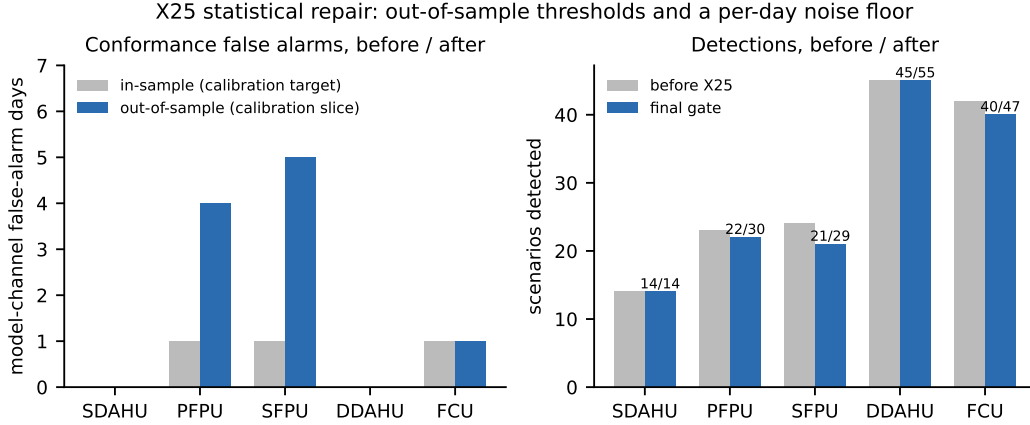


Figure 6: The statistical repair. Left: the conformance channel’s false-alarm days as a calibration target (grey) and out-of-sample after the three-way split (blue). Right: detections before and under the final calibration; the per-day residual null removed seven residual-only detections, one offset by a new channel.

Section 6.10) (on the cooling coil the recorded water-side temperature difference, valve-open time and flow stay within about three per cent of the fault-free year) and the 20 % outdoor-air damper leak; the per-day residual null of the statistical repair (Figure 6) then took back both the 20 % leak, which the outdoor-air-fraction residual had caught, and the 50 % leak, which had been detected at onboarding, so both leaks were misses until the coverage audit recovered them.¹³ What the fifth system adds is not the row but the failure: a semantic constant in a scoring script is a hypothesis about the equipment class, and two artefacts that compute one quantity two ways will agree until the first class on which they do not.

6.10 The coverage audit: every recorded column, one system at a time

The misses that remained after the statistical repair were read as limits of what the datasets record. A pre-registered envelope experiment (X32, experiments being numbered as in Appendix A: a healthy time-of-day envelope per signal, with sustained departures from it emitted as events) was withdrawn before any computation when a read-only scan of its design against the data showed that a third to a half of all signals leave such an envelope on healthy holdout days; the same scan found that for six of the fourteen misses it classed as reachable from a recorded column the decisive signal was a column no configuration mapped (the audit that followed recovered sixteen, several of them fouling files the scan had classed as unreachable): the configurations had been written from the control sequences, not from an audit of every recorded column. A pre-registered series (X33a–e) audited every column of every system in the project’s onboarding order, mapped every usable one, gave each a physically named reader (a residual, a rule, a gate or an oscillation signal) and excluded the rest with a written reason that a build-time guard now enforces. Every band came from the fault-free file; the state alphabet was not touched, so the process-mining rows had to regenerate byte-identically; and a pre-registered adoption rule removed any new channel that was credited nowhere and cost more than one held-out day (none was).

What the recovered misses were, and which physics carried them (Table 8): airside reheat-coil

¹³outputs/x19_onboarding_run1.json, x19_onboarding.json, benchmark_v6_fcj.json, union_fpr_fcj.json; docs/plans/2026-09-24-x19-fcj-onboarding-prereg.md with Amendment 1; tests/test_artefact_consistency.py.

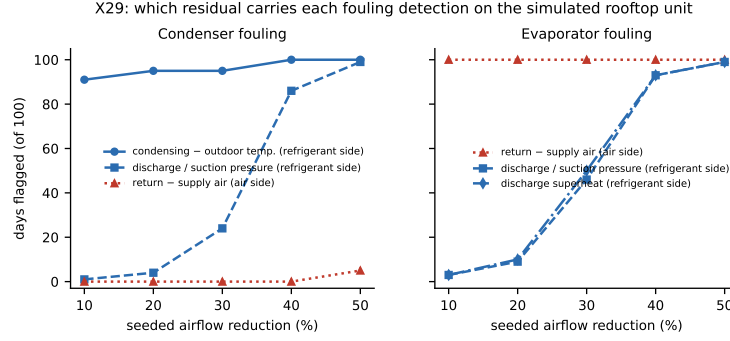


Figure 7: Which residual carries each fouling detection on the simulated rooftop unit, by seeded airflow reduction (days flagged of 100, from the per-residual credit artefact). Condenser fouling is carried by the refrigerant-side condensing-temperature approach at every severity; evaporator fouling is carried by the air-side return-minus-supply residual at every severity, with the refrigerant-side residuals joining from 30 %.

System	columns mapped	detected, before → after	false alarms, before → after	lost
SDAHU	12 → 26 of 30	14 → 14 / 14	1 → 1	0
PFPU	56 → 109 of 109	22 → 26 / 30	7 → 8	0
SFPU	56 → 109 of 109	21 → 25 / 29	6 → 9	0
DDAHU	79 → 114 of 114	45 → 50 / 55	3 → 3	0
FCU	29 → 29 of 29	40 → 43 / 47	4 → 5	0
all	232 → 387 of 391	142 → 158 / 175	21 → 26 of 480	0

Table 8: The coverage audit. Sixteen scenarios recovered, none lost, five false-alarm days added over 480 held-out days, the model and device rows identical on every system (the log hashes change, because the new readers emit signature events). Totals use the naive single-duct count. Ledgers outputs/x33_coverage_*.json; per-rule credits outputs/x33_residual_credits_*.json.

fouling on both fan-powered units, through the series fan’s power per cfm of air delivered (“same power, less air”) and, on the parallel unit, the relationship between two airflow sensors; the fan-flow restriction, through the same fan-power indicator; the room-temperature sensor biases, through the return air falling away from the biased zone’s reading; the hot-deck static and temperature biases on the dual-duct unit, through the unbiased copies of both quantities the mixing boxes record (exact, 282 of 282 days, at zero false alarms); one waterside fouling, through the pump’s bypass flow; and the two damper leaks on the fan coil unit, through the outdoor-air fraction of discharge flow at a fixed damper position. Two of the recoveries deserve a caveat the ledgers carry: the fan coil unit’s moderate heating-waterside fouling is seen only because the controller drives the fan to a speed the healthy year never used, an operating-point indicator rather than a coil measurement; and one predicted recovery on the parallel unit’s waterside fouling is visible in the data yet uncertifiable, because the water flow only reveals it on the ten days a year the reheat valve saturates, too few to estimate a null.

Two findings of the series are about the protocol rather than the detector. The first step exposed a drift between the importable library and the benchmark scripts (Section 5); the second exposed that the fan-powered units’ documentation names the wrong zone for every seeded fault (erratum E7, Section 5.5). The seventeen misses that remain are coil fouling on every system, whose recorded values sit inside every healthy band on every column. The claim the series licenses is narrow: on these datasets, the remaining misses are limits of what is recorded and how often,

not of the detector’s vocabulary or its gate, and the earlier reading, that all thirty-three were, was wrong for sixteen. Two disclosures bound the claim: which columns to read was decided after a read-only scan of the fault files had shown which columns separate which misses, so the readers were fault-informed even though every threshold was blind; and five of the sixteen recoveries sit less than one band width outside the healthy extremes, so a different healthy year could move the band edges they clear. Two later pre-registered tests bound that second disclosure: all five edge recoveries survive bands about ten per cent narrower, and four of the five survive injected field conditions (Section 8). The series therefore shows that these misses were reachable from the recorded columns with blind thresholds, not that a blind onboarding would have reached them; the blind onboardings of Section 6.9 remain the honest prior for a new building.

7 What the Evidence Ruled Out, and What It Settled

Thirty pre-registered experiments produced ten results that went against the hypotheses they tested. This section reports them, because they settle two things the confirmations cannot: what the discovered process models contribute, and how far the method reaches on the data it has. None of them removes a detection from Section 6; each removes a claim about why a detection happens or where the method’s reach ends. Table 9 lists the ten with what fired and what survived; the paragraphs that follow narrate the ones that carry the argument.

Table 9: Ten results that went against the project. Rows 4 and 5 are reported in Section 5, row 8 in Section 6.7 and in the last paragraph of this section, row 9 in Section 6.7, row 10 in Section 6.9 and in the table itself; the rest in the text that follows.

#	Hypothesis or result	Refutation condition / what fired	What survived
1	Hypothesis: discovered models out-detect hand-written rules.	A one-line matched rule (occupied workday and AHU heating never active) covered 192 days against the model channel’s 135 when the falsifier fired (178 in the current artefact, after the threshold repairs) on the same scenario, with zero false alarms on 365 healthy days. The falsifier fired.	The models were design-time aids: an invariant surfaced during discovery, the expert wrote the rule, and thereafter the models contributed no detection. The matched rule is kept in every scorecard as a control.
2	Hypothesis: behavioural structure discovered on one building supports detection on another without any configuration for the second building.	The detection hypothesis failed its pre-registered falsifier, yielding zero to one detection per building pairing under the stricter of the two pre-registered gates (the looser one gives 0, 1 and 5).	Real shared structure: fan-powered-unit logs score 0.15 and 0.18 higher in mean fitness on the single-duct net than shuffled versions of themselves. Portability is claimed at configuration level only.
3	Result: the conformance channels detect faults no other channel catches.	Repeated audits found that on alphabets whose events repeat many times a day the discovered nets accept activities in almost any order, and every genuine model-channel detection was a frequency deviation surfacing through alignment cost.	The frequency channel, which measures those deviations directly and in both directions; the conformance channels had nothing left that was uniquely theirs.

Table 9 (continued)

#	Hypothesis or result	Refutation condition / what fired	What survived
4	Result: the single-duct system scores 14 of 14.	An audit established that the dataset’s fault-free file was simulated on a different configuration branch than every fault file; under a branch-corrected band, one scenario’s only deployed channel produced zero flags on the 148 days it could be evaluated.	13/14 adjudicated, quoted together with 14/14 naive, and the standing rule that healthy-versus-fault divergence is fault evidence only after a fault-versus-fault control.
5	Result: a day-level significance test adopted early in the work.	The test rested on an impossible null (the discordant cell it tested was zero by construction), so it awarded significance to noise.	A binomial against each channel’s own measured holdout rate, after thirteen scenario rows on the parallel unit lost their significance marks. The standing rule is that the null is always the channel’s own noise.
6	Hypothesis: healthy silence tuned at one-minute sampling is preserved at fifteen-minute sampling.	Configurations tuned at one-minute sampling gain 28 to 64 healthy days with a spurious signature-rule firing at fifteen-minute sampling and one residual coverage drops by more than a quarter; four falsifiers fired. A separate pre-registered measurement of a known property is not counted here because it confirmed its hypothesis: a single score-carrying contaminated day removes the two negative-bias scenarios’ detections entirely (from 164 of 164 days to none), by dragging the band edge to -7.19 .	Contamination from a mild source announces itself on 100% of the contaminated days (5/5, 13/13 and 27/27 signature days), whereas the dangerous source is visible on none of its 27. The transfer claim was scoped so that the healthy-silence gate is re-run at the deployment sampling rate.
7	Protocol failure: a pre-registered test of whether discovery predicts where a rule transfers.	The model quantity is constant within a building, so the nine test cells were three buildings counted three times each; the valid test permutes buildings, and with three the smallest attainable p is 0.167 against a criterion of 0.05. The test never had power, and the blinding covered only the four cells whose rules were fitted on the same year they were scored on; three dated amendments record it.	No conclusion in either direction. The lesson: pre-registration did not fix the unit of analysis.

Table 9 (continued)

#	Hypothesis or result	Refutation condition / what fired	What survived
8	Hypothesis (fault-informed, disclosed): a coil-effectiveness residual, water-side ΔT per gallon, separates reheat-coil fouling at day level.	Significant on one of twelve fouling scenarios, already detected by other channels; nothing newly detected; falsifier F-X13.b fired. The statistic's day-median distribution overlaps healthy almost entirely (bands 0.6–94 °F/GPM) because the ratio explodes at low flow.	The fouling misses need a different sensor, not one more residual: a pre-registered coil heat-transfer estimate gained nothing, eight of the ten missed fouling scenarios carry no signal in the event log at the deployed alphabet, and on LBNL's simulated rooftop unit, the one dataset that records the refrigerant side, the unchanged detector catches every fouling severity (10 of 10, monotone) at 0 false-alarm days in 28. The evidence and its limits are given in the last paragraph of this section.
9	Hypothesis: the discovered model was starved of vocabulary, healthy-derived actuator and flow band states let the state channels detect within the false-alarm budget.	One missed scenario becomes significant (a zone-damper signature that grows with fault severity, checked fault against fault) and the single-duct model channel fires on 14 scenarios instead of none, at 7, 18 and 21 false-alarm days in 96; falsifier F-X14.a fired. Alignment-based conformance on the enriched fan-powered logs was computationally impractical and not evaluated (three recorded amendments).	Vocabulary was part of the limit; richer vocabulary buys detections at false alarms the budget forbids. Whether a stricter calibration keeps the gain inside the budget is untested.
10	Test: the transfer test re-run with four scorable buildings, pre-registered with the decision rule that the model quantity equals the log's own heating-day count.	The equality holds on three buildings and fails on the fourth only because the miner's noise filtering dropped the heating-off activity; scored as pre-registered, the test passes (Spearman $\rho = -1$, exact one-sided $p = 1/24 = 0.042$). A control added afterwards and not pre-registered, the raw heating-day fraction with no discovery in it, orders the buildings identically and passes identically; two falsifiers fired (the equality, and the claim that the net makes heating obligatory).	A formal pass that the paper declines to read as evidence about discovery, because the discovery-free control passes too. A variant that cannot be circular (a net fitted on one building reading another's fault-free year) was then pre-registered and run: it is not the count, and as a predictor it orders the four buildings no better than the count does. Reported as a protocol failure, not as a result in either direction.

Discovered models do not out-detect hand-written rules. On the series fan-powered unit, zone damper-stuck and negative airflow-bias faults erase the unit's daily heating rhythm, and the discovered healthy net flags the absence. A one-line matched rule (occupied workday and AHU heating never active, with zero false alarms on 365 healthy days) covered 192 days against the model channel's 135 on the same scenario (178 in the current artefact, after the threshold repairs; the falsifier

fires either way). The 135 is a threshold-based count and should be read alongside the count that survives replacing the interpolated threshold with the most conservative one, 22 to 26 days per scenario, since roughly four fifths of the threshold count is attributable to a threshold interpolated from a single holdout day. The falsifier fired. An invariant surfaced during discovery and the rule was written afterwards; whether the model could have predicted where that rule would hold is the question the seventh entry could not test. The matched rule is now kept in every scorecard as a control. This refutation is why the paper does not present the models as a detector.

Behavioural structure does not transfer without configuration. The cross-system grammar was pre-registered with the hypothesis that structure discovered on one building would support detection on another. The grammar analysis did find real shared structure: fan-powered-unit logs score 0.15 and 0.18 higher in mean fitness on the single-duct net than shuffled versions of themselves, while the other four cross-system cells are null (the difference from shuffled lies between -0.005 and 0.000), so only the single-duct net is order-sensitive. But the detection hypothesis failed its pre-registered falsifier, yielding zero to one significant detection per building pairing. Portability is therefore claimed at configuration level only, which is what the onboarding evidence supports and no more.

The models' detections were count deviations in disguise. Audits of the device-stratum results reached one conclusion repeatedly: on alphabets whose events repeat many times a day the discovered nets accept activities in almost any order, and every genuine model-channel detection was a frequency deviation (extra heating episodes, collapsed fan cycles, a vanished daily rhythm) surfacing through alignment cost. The frequency channel was built to measure those deviations directly and in both directions, and once it existed the conformance channels had nothing left that was uniquely theirs, which the ablation above confirms.

A pre-registered test that never had power. After the ablation one claim remained: that the models locate invariants, so that a rule written from one would hold wherever the model's support for the invariant is high. A test was pre-registered, with the model-derived quantity, the predictions and the falsifiers committed before computation and the quantity computed by an agent forbidden from opening the rule-firing artefacts. Two things went wrong, and both are recorded in dated amendments. The seal was narrower than intended: the pre-registration's own disclosure section printed the transplanted rule's healthy firings, so blinding held only for the cells whose rules were fitted on the same fault-free year they were scored on, where near-zero firings are arithmetic rather than transfer. And the test never had power: the model quantity is a property of a building's net and is constant across rules within a building (0.000 , 0.455 and 0.978), so the nine test cells were three buildings counted three times each; the valid test permutes buildings, and with three buildings the smallest attainable one-sided p is $1/3! = 0.167$ against a criterion of $p < 0.05$ (with the two buildings the amended test actually scored, $1/2! = 0.5$). The reported $p = 0.20$ is therefore not a measurement, and the paper draws no inference about the claim in either direction. What the exercise did show is reported as observation: the series unit's net treats heating as obligatory (356 of its 364 healthy daily variants contain the activity) while the parallel unit's net treats it as optional, and the alignment-based support equalled the raw log count exactly on every system, so at the unit stratum the net's reading added nothing over counting. The lesson is the entry's contribution. Pre-registration fixed the quantity, the predictions and the falsifiers, and two amendments written under the same discipline did not fix the unit of analysis; a power calculation on the wrong unit survives pre-registration intact. The remedy adopted is a pre-registration field naming the unit of replication and the minimum attainable p under a permutation of that unit, computed before any prediction is written. A powered test of the transfer claim needs more buildings than three; the four-building re-run (row 10) is reported in Section 6.9.

Calibration is fragile to contaminated training data. A band set from the extremes of the training days can be moved by a single day, and one contaminated day that carries a residual score

in the fault-free year is sufficient to remove the two negative-bias scenarios' detections entirely (from 164 of 164 days to none), by dragging the band edge to -7.19 . The finding must be quoted with its detectability split or it misleads in either direction: contamination from a mild source announces itself on 100% of the contaminated days (5/5, 13/13 and 27/27 signature days) and is therefore caught, whereas the dangerous source is visible on none of its 27 days. A related test found that configurations tuned at one-minute sampling gain 28 to 64 healthy days with a spurious signature-rule firing at fifteen-minute sampling; four falsifiers fired, and the transfer claim was scoped so that the healthy-silence gate is re-run at the deployment sampling rate rather than transplanted.

The fouling misses need a different sensor, not one more residual. A pre-registered coil heat-transfer coefficient (UA, from water-side flow and the four terminal temperatures) later gained no fouling scenario on the dual-duct or fan coil unit and cost two false alarms on the latter; eight of the ten missed fouling scenarios carry no signal in the event log at the deployed alphabet, and one of those eight becomes detectable on an enriched one (Section 6.7). The positive control: on LBNL's simulated rooftop unit, the one dataset that records the refrigerant side, the unchanged detector catches every condenser- and evaporator-fouling severity down to 10 % (10 of 10, monotone) at 0 false-alarm days in 28. A per-residual credit artefact (Figure 7) says which physics carried it: condenser fouling by the condensing temperature's margin above outdoor air (the approach), a refrigerant-side point no air handler records, on 91 to 100 of 100 days at every severity; evaporator fouling by return-minus-supply air, an air-side residual, because this dataset seeds both fouling types as fan airflow reduction, and the refrigerant-side residuals see the evaporator only from 30 %. The evidence that refrigerant-side points make fouling observable is therefore the condenser half; the 17 fouling misses that remain across the four air-side datasets after the coverage audit (Figure 2) are simulated as coil-conductance changes under supply-temperature control, a different mechanism, and this test does not show that refrigerant points would recover them.¹⁴

8 Limitations

The false-alarm budget is the property that has been stressed hardest, and it held: under injected sensor error, change-of-value logging and schedule shifts on the shipped configurations of all five simulated systems, the worst case is 7 of 96 held-out days, each condition costs at most one detection per system, and a narrower calibration gains nothing. The limitations that remain are these.

Every detection result is simulated, and simulation flatters. Noise-free years make zero-false-positive thresholds cheap, bands thin and silence easy. Four pre-registered tests bound that flattery without removing it. Under per-sample jitter at building-sensor precision (0.5°F , 2 % of flow, 0.01 of travel) the detection counts of the time were unchanged on the three original systems and false alarms stayed at 1 to 4 days in 96, though which marginal scenarios were detected shifted (two lost and two gained on the parallel unit).¹⁵ Under field sensor error with 5 % feedback quantisation, change-of-value logging with stale repeats and gaps, and schedule shifts with holidays, each and all together, applied to the fault-free year and every fault file and re-fitted, the budget held on every system both before the coverage audit (worst 3 of 96 on three systems) and after it on all five (worst 7 of 96, the parallel unit under the schedule shift); each condition cost at most one detection per system, named in the artefacts, and nothing was gained.¹⁶ One run exposed how a band set from

¹⁴outputs/x29_residual_credits.json; X29 Amendment 1 in docs/plans/2026-09-24-x29-refrigerant-observability-prereg.md: the closed artefact's one false alarm was a model and oscillation flag on the record's final date, which holds a single 00:00 row; dates with under 720 rows are now dropped at conversion and the regeneration changed no detection.

¹⁵outputs/x9_x10_robustness.json.

¹⁶outputs/x31_field_conditions.json, outputs/x35_field_conditions.json; pre-registrations of 2026-09-24 and 2026-

the extremes of the training days can move on a single day: a schedule shift turned the one training day that set the top of a hot-deck band into a holiday, the band narrowed, and three sensor-bias detections appeared at no false alarm, a prediction failure in the detector’s favour that is reported and not credited. An alternative calibration at the 2nd and 98th percentiles, about ten per cent narrower, gained no scenario, lost none, kept all five recoveries that sit within one band width of the healthy extremes, and added thirteen false-alarm days over 480; it fired its budget falsifier and the deployed calibration is unchanged.¹⁷ The one measured building (Section 6) supports a false-alarm rate, 6.1 % of held-out days, and nothing more; its single fault case was not visible in its recorded signals. Until the deployment phase is complete the framework is validated in the laboratory, not in the field, and a real building’s baseline year is not fault-free by construction, which the contamination result of Section 7 makes concrete.

Two reporting choices were made after their effect was known, and are disclosed. First, the deployed false-alarm rates treat the seasonal rate channel as corroboration that may not alarm alone. With every channel counted the single-duct system false-alarms on 15.6 % of held-out days, nearly all of it from that channel; demoting it was adopted after that figure was known, was verified to cost no detection and no delay, and both figures are published together. A check that splits the holdout in two passes on the single-duct and parallel systems and fails on the series system, where the noisiest channel differs between halves and is not the demoted one. Readers comparing against a method that reports every channel should use 15.6 %. Second, until the three-way calendar split of Section 4 the conformance channels’ false-alarm rows were calibration targets rather than measurements, because their thresholds were fitted on the same held-out days those rows report. The split measured that debt: the model channel’s out-of-sample rows on the fan-powered units are 4 and 5 of 96 against 1 and 1 in-sample, and the deployed budgets moved from 5.2 and 4.2 % to 7.3 and 6.2 %. Making the residual channel’s null a per-day rate on its own held-out days then removed seven residual-only detections, one offset by a new channel; the loss fired a pre-registered falsifier and is reported as a detector change. Every reported row is now out-of-sample, though every threshold still comes from the same fault-free year.

No correction is applied across the many significance tests. About 1,600 binomial tests are run at $\alpha = 10^{-3}$ (eight channels by 199 scenarios on six systems, the device channel per terminal unit), so under independence the chance of at least one false cell somewhere in the study is near 0.8 and about 1.6 false cells are expected; per-day observations within a scenario are also correlated, so every p is optimistic. Correction is applied only across devices within the device channel. Confidence intervals are given for the false-alarm rates and detection counts, and all are single-run figures on one simulated year per system.

The benchmarks do not contain the fault class the introduction argues from. The seven datasets seed component, sensor and control-loop faults: stuck and leaking dampers and valves, sensor biases, coil fouling, airflow and refrigerant-line restrictions, refrigerant charge, unstable loops and a reversed controller. None seeds a schedule, override or setpoint-reset fault, the behaviour-based class that dominates the field survey the introduction cites. The case for sequence-aware detection on that class is motivated here, not tested; the detections reported are of faults a rule library could in principle enumerate, and Section 6.4 measures how many of them one such library does.

Reach. Gradually developing faults are detected late (the 114-day tails of Section 6); the comparison against deployed practice covers three systems and one rule library; and six simulated systems across five equipment classes demonstrate configuration-level portability without establishing generality across commercial HVAC equipment. A collection of eight further labelled air-handler fault

09-27.

¹⁷outputs/x36_quantile_band.json.

datasets published in 2026 [24] is the obvious next target.

9 Conclusion

STRATA works on the benchmarks it was tested on. A stratified event log built from the control sequence, process models discovered at device and unit level, and eight calibrated, physically interpretable channels detect 158 of 175 seeded faults on five public benchmark systems (157 once one dataset defect is adjudicated), most on the first day, inside a stated false-alarm budget of 26 days in 480 held-out fault-free days. The alarms name the correct terminal unit in 37 of 50 checkable detections and a wrong one in none, name the faulted subsystem in 83 % of all detections, and each carries the evidence behind it, which a scalar anomaly score of equal detecting power cannot supply. The budget held under injected sensor error, change-of-value logging and schedule shifts on all five systems (worst 7 of 96 days) and on one measured rooftop unit (3 of 49). Against the rule battery that commercial building automation systems run today, wired correctly and held to the same gate, STRATA is level on the air handler and detects three times as many faults on the terminal units, because it reads the per-zone sensors the rules ignore; the pre-registered claim to beat current practice everywhere was withdrawn, and the comparison is stated as the evidence supports it.

Process mining had not previously been applied to fault detection in building HVAC systems. This work made the application and measured what it contributes. Process mining helps to get detection but is not detecting on its own: the strata and the event abstraction give every alarm its location, the log’s count channels are the only ones that see the unstable-controller faults, and invariants that discovery surfaced became several of the rules that detect; conformance checking of the discovered models, tested as the detector, caught nothing the calibrated channels missed, and removing it changes no result. That division of labour is the paper’s answer to the question that motivated it, and it was settled by pre-registered tests rather than assumed.

What this is worth to a building operator is concrete. The false-alarm budget is a measured property rather than a promise: it held on a real rooftop unit, 182 fault-free days over two summers with no schedule point, and on every simulated system under the sensor error, stale logging and moved schedules that field data impose. Onboarding is a configuration file rather than a software project: the second system type took about 1.5 hours by commit timestamps, the third was derived from it in minutes, and the fourth and fifth, a dual-duct air handler and a fan coil unit, were onboarded blind under a pre-registered protocol with no change to the pipeline code. An alarm names the zone and the subsystem, which coil, which fan, which damper section, which zone’s mixing box, and states the physical relationship that broke and by how much. The seventeen faults still missed are coil fouling whose recorded values sit inside every healthy band on every column, a limit of what the datasets record rather than of the detector; the audit that established this, which maps every recorded point or excludes it with a written reason, is the checklist a deployment starts from.

The proposal that began this work named Thompson Rivers University’s campus energy system as the intended testbed: a low-carbon district energy system built with Creative Energy and BC Hydro and switched on in September 2026 to serve the campus’s buildings [71, 72]. The building-side equipment there is of the classes tested here. A 2022 ASHRAE Level 1 energy study of the Campus Activity Centre lists an air-handling unit serving variable-air-volume boxes, fan coil units and five rooftop units, with every major system except the heat pumps serving nine of the fan coils on an Automated Logic building automation system that already records the trend data STRATA reads, and with the building in the second phase of the plan for connection to the district plant [67].

Deploying STRATA there is the next step and is not done: every detection result in this paper was obtained on benchmark data, and the one measured building contributes a false-alarm rate, not a detection rate. Deployment needs three things. A fault-free period of trend data at the site’s own sampling rate, audited for contamination, because a single faulty training day can move a band edge (one year on the benchmarks; 182 days sufficed to measure the rooftop unit’s budget). A sensor map written from the point list as a configuration, with every recorded point mapped or excluded with a reason. And a pass confirming that the rules are silent on that fault-free period at that sampling rate before the detector scores a day, because configurations tuned at one rate do not transfer to another. What it would deliver is a per-day, per-unit alarm stream whose false-alarm rate has been measured on the site’s own held-out days before the first alarm is trusted, each alarm naming the unit, the subsystem and the relationship broken. The district plant itself, heat pumps and a distribution network, is an equipment class this work has not onboarded; its configuration would be written the same way, and its silence pass would be the first evidence of whether the calibration holds there.

Scaling follows the same path, and it has been walked: six datasets beyond the reference one, two fan-powered units, a dual-duct unit, a fan coil unit and two rooftop units, went through the same pipeline code, each at the cost of a configuration. What travels with the configuration is the discipline: the silence gate that must pass before a configuration scores anything, the regeneration gate that fails the build when a published number moves, and the pre-registration form that fixes predictions and falsifiers before the first fault file is opened. Two limits travel too: configurations are sampling-rate specific, so the silence gate is re-run at each site, and a blind onboarding’s first detection count (45 of 55 and 41 of 47 on the two systems onboarded that way) is the honest expectation for a new building before its columns are audited. The eight further labelled air-handler datasets published in 2026 [24] are the next external-validity target before a campus.

The project’s name, ProcessHeal, states the destination: a building that detects its own faults, locates them, diagnoses them and corrects them. This paper fixes where the work stands on that path. Detection under a measured budget, localisation to the zone and the subsystem, and an alarm that states its evidence exist. Diagnosis was tested: a pre-registered layer that reads the sign of every significant residual, applies a consensus rule over redundant copies of one quantity and votes on the nearest fault pattern names the fault family in three of four resolved cases (85 of 114, with 44 unresolved) and resolves the five dual-duct damper cases the carrying rule had placed at the wrong end of a redundant sensor pair, but misses both of its pre-registered bars and is reported, not adopted. What it showed is a boundary of passive observation: a stuck actuator and a biased sensor on the same loop, or a fouled coil and the fan working harder against it, leave the same signed footprint in the recorded points. A self-healing loop therefore needs three things this work does not yet have: active tests that perturb a device and watch the response; a witness for cause against response, whether a second sensor or a commanded excursion; and an operator study before any automated action, because writing a correction back to a building’s controls is a field problem in its own right, solved so far by hand-coded algorithms for enumerated faults [44,63], and process mining has been proposed as an enabler of self-healing in the Internet of Things [68]. This is a research programme and is stated as one; the parts of it that already exist are the detector, the localisation, the coverage audit and the protocol under which each further step would be pre-registered.

The contribution this paper is surest of is that protocol: pre-registration with binding falsifiers, independent recomputation before any number is quoted, and automated tests that pin published values, fired refutations and disclosure obligations. Its evidence is what it did to the project’s own claims: ten pre-registered results went against the project and four later falsifiers fired, all published; eight defects in a widely used public benchmark were documented with reproducible evidence; and an adversarial audit voided a completed comparison against commercial practice

twice rather than let it be published, so that the version quoted is the one that survived. In a subfield where independent verification is rare [52], a result that has survived a genuine attempt at refutation is worth more than a larger number that has not, and a detector whose budget, location and onboarding cost are measured is one an operator can hold to account, which is what the introduction asked for.

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A Every pre-registered experiment

Table 10 lists every pre-registered experiment in the order it was run, with its pre-registered prediction, its outcome and what fired. Each row resolves to a ledger under outputs/ or, where noted in the repository’s research log, to the phase report; amendments are folded into the row they amend.

Table 10: Every pre-registered experiment, its prediction, its outcome and what fired.

ID	Question	Pre-registered prediction	Outcome	Fired / failed
—	Do discovered unit models out-detect a hand-written rule?	The model channel covers more fault days than any matched rule	A one-line matched rule covered 192 days against the model channel’s 135 (178 in the current artefact), at 0 of 365 healthy false alarms	Falsifier fired (matched-rule ablation)
—	Does behavioural structure transfer zero-shot to a new building?	Source-fitted bands detect faults on the target system	0 of 14 single-duct scenarios detected under the confidence-bound gate	Falsifier fired (cold-start detection)
—	Does model support predict where a transplanted rule holds? Three amendments.	Spearman $\rho < 0$ with permutation $p < 0.05$ over nine cells	$\rho = -0.69$, $p = 0.200$; the valid null over three buildings has minimum $p = 0.167$	F1 and F2 fired, then both voided; never had power
X5	Are severity ladders monotone in dose?	No ladder is significantly anti-monotone	Series-unit fouling $\rho = +0.87$; no anti-monotone ladder; F-X5.a nearly unfireable at $n = 5$	None
X7	Does healthy silence survive 15-minute resampling?	Healthy signature days and residual coverage are preserved	Healthy years gain 28, 64 and 64 signature days; one residual’s coverage falls 31 to 1	F-X7.a (three systems) and F-X7.b
X8	Does training contamination break extreme-value calibration?	Breakdown at small contamination; mild sources stay visible to rules	One score-carrying contaminated day collapses both negative coil-bias rungs, 164 of 164 to 0	None
X9/ X10	Is the deployed detector robust to sensor jitter and to the holdout split?	Detections change by at most 3; false alarms under 10 %	Counts unchanged at $1 \times$ jitter, within one on the mirrored split; 0 to 4 false alarms of 96	None

Table 10 (continued)

ID	Question	Pre-registered prediction	Outcome	Fired / failed
X11	Is the single-duct OA-bias detection the fault or the configuration branch?	Five baseline estimators agree; the branch offset explains the flags	Offset +1.06 °F; OA-bias drops to 0 flags on 148 window days; 14/14 becomes 13/14	F-X11.d fired on run 1; leg re-specified
X12	Does the model’s time perspective recover missed scenarios?	The duration channel newly detects missed scenarios	11 scenarios significant but 0 newly detected; unique against frequency on 11	None; P3 failed without one
X13	Does a water-side coil-effectiveness residual separate reheat fouling?	At least 6 of the 9 fouling scenarios it counted significant	1 of the 12 fouling files scored significant, already detected elsewhere; nothing newly detected	F-X13.b
X14	Was the discovered model starved of vocabulary?	At least one newly detected scenario inside the false-alarm budget	One newly detected; the single-duct model channel fires on all 14 scenarios instead of none (5 over the three systems before); 7, 18 and 21 false alarms of 96	F-X14.a (budget)
X15	Does the enriched gain survive frequency-only scoring under two splits?	The gain is significant under both splits and inside budget	The same one scenario is significant under both splits; 41 significant against 12 deployed	None
X17	Can a dual-duct air handler be onboarded by configuration alone?	Config-only, gates clean, at least 60 % of scenarios detected	45 of 55 detected (81.8 %) at 3 false alarms of 96; one silence iteration	None; P6 failed (5 fouling, not ≤ 4)
X18	Does the enriched-frequency gain replicate on the dual-duct unit?	A missed fouling scenario becomes significant under both splits	0 newly significant; enriched 22 against deployed 8 on already-detected scenarios	F-X18.b
X19 run 1	Can a fan coil unit be onboarded by configuration alone?	Gates clean, at most 10 false alarms of 96, no conformance-only scenario	All 47 scenarios “detected” at 28 false alarms of 96; 6 conformance-only	F-X19.b and F-X19.d
X19 A1	Does the amended fan coil onboarding hold?	Budget restored and no conformance-only detection	41 of 47 detected at 4 false alarms of 96; 0 conformance-only	None
X20	Is model support the log’s own heating-day count across four buildings?	The identity holds on every building; $\rho < 0$ at $p < 0.05$	$\rho = -1$, exact one-sided $p = 1/24 = 0.042$; the identity misses by 44 days on the fan coil unit	F-X20.a and F-X20.b
X21	Can a net’s reading of a foreign year escape being that log’s count?	The bound holds, the tautology persists, the ordering matches the count	The tautology fails on 6 of 12 pairs; the foreign predictor gives $\rho = -0.6$ against the count’s -1	F-X21.a; P3 failed without one
X22	Does the conformance channel fire on injected order violations?	Reversal is significant on the single-duct unit	Significant on 2 of 5 systems; the single-duct hit rate is 0 at zero false alarms	None; P1–P4 failed
X24	Does an outdoor-air-fraction residual add detections?	The 20 % damper leak is detected, budget held, nothing lost	Fan coil 41 to 42 of 47 at 4 false alarms of 96; no system loses a detection	None
X25 steps 1–2	Are the conformance thresholds calibrated out of sample?	Rows out of sample, budget ≤ 10 of 96, at most 3 verdict flips	All held at 0 verdict flips; the in-sample rows had understated false alarms by 3 and 4 days	None

Table 10 (continued)

ID	Question	Pre-registered prediction	Outcome	Fired / failed
X25 step 3	Does a $\max(\text{fp}, 3)/n$ floor on the pooled denominator keep verdicts?	Counts within 2 and at most 3 verdict flips	Held at 2 verdict flips; one dual-duct scenario gained, one series-unit scenario lost	None
X25 step 4	Does a per-day residual null keep the detections?	Each system’s count stays within 2	The dual-duct unit loses 3 detections (the series unit 2, the parallel and fan coil units 1 and 2) and 8 verdicts flip across the five systems; budgets unchanged (step-level artefact, before the fan-law channel; the final regeneration, <code>x25_statistical_repair.json</code> , nets seven lost and one gained on the dual-duct unit, the count Section 8 quotes, and fires F-X25.c on the series unit)	F-X25.c
X26 (+A1)	Does the false-alarm budget hold on a real building?	At most 3 silence iterations, false alarms under 10 %, the fault case flagged	6.1 % false-alarm days with every channel in budget; the 40 % undercharge case is not flagged	None; both case predictions failed
X27	Does a fan-law virtual static recover static-pressure biases?	At least 7 of 8 static-bias scenarios detected	6 of 8 detected; budget held and nothing lost	F-X27.b
X28	Does a water-side coil UA channel see the simulated fouling?	At least 3 dual-duct and 2 fan coil fouling scenarios gained	0 gained on either system; run 1 also broke the fan coil budget	F-X28.b; run 1 also F-X28.a
X29	Is fouling visible when the refrigerant side is recorded?	At least 3 of 5 condenser and 3 of 5 evaporator severities detected, monotone	All 10 fouling severities detected and monotone at 0 false alarms of 28	None
X30	Does any other process-mining instrument see the deployed misses?	No cell detects a miss within 3 false alarms	33 cells against 33 missed scenarios, 0 detections; worst cell 7 of 72 (9.7 %)	F-X30.b (two cells not evaluated)
X31 (+A1)	Does the detector survive field noise, change-of-value logging and schedule shifts?	Budget held everywhere; the schedule arm within ± 1 detection	Budgets held (run before the coverage audit); the dual-duct schedule arm reaches 48 against the clean arm’s 45	None; P3 and the A1 prediction failed
X32	Does a healthy time-of-day envelope per signal, with departures as events, carry detection?	Envelope counts inside the budget; the envelope does not see fouling; order among deviation events adds nothing beyond their counts	Withdrawn before any computation: a read-only design scan showed a third to a half of signals leaving the envelope on healthy holdout days (9–19 % of healthy days flagged); no artefact	None; withdrawn unrun, replaced by X33
X33a-e	Does mapping every recorded column recover the misses? (one system at a time)	No loss; budget $\leq$ clean + 3; named recoveries per system	142 $\rightarrow$ 158 of 175, none lost, 21 $\rightarrow$ 26 false-alarm days of 480, model and device rows identical; PFPU and DDAHU fouling recovered against prediction	None; three predictions failed upward, one (PFPU waterside) downward

Table 10 (continued)

ID	Question	Pre-registered prediction	Outcome	Fired / failed
X34	Does the carrying rule name the seeded component?	Exact $\geq 60\%$; exact or subsystem $\geq 85\%$; wrong subsystem $\leq 10\%$	Exact 109 of 158 (69%), subsystem 22, wrong 20 (12.7%), unnamed 7	F-X34.a; the component-level claim withdrawn
X35	Do the field conditions hold on the post-coverage configurations of all five systems?	Budget $\leq$ clean + 3; noise and COV within -4 ; at least two edge recoveries lost under noise	Budget worst 7 of 96; at most one detection lost per system and condition; one edge recovery lost; clean arm equals the scorecard	None; P3 failed
X37 (3 runs)	Do signed patterns and redundant-pair consensus diagnose family and component?	Family $\geq 80\%$ with $\leq 15\%$ unresolved; wrong subsystem $< 10\%$	Family 74.6% with 27.8% unresolved; wrong subsystem 10.1%; the five dual-duct damper cases resolved	F-X37.a, F-X37.b
X36	Do percentile bands keep the edge recoveries and stay in budget?	False alarms rise ≤ 3 ; the five edge recoveries kept; ≥ 2 new detections	Nothing gained or lost; all five kept; false alarms +4 on both fan-powered units	F-X36.a
X38 (+A1)	Does the repaired Guideline 36 battery match STRATA under the same gate?	STRATA at least as many at lower false alarms (three framings after A1)	Silent-rule battery 12 of 14 at 1 vs 13 at 1; 8 of 30 and 9 of 29 at 0 vs 26 and 25 at 8 and 9	F-X38.a under G2, G3 and S

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